EE/CE 252/251
Signals and Systems

with

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@

Habib University

Signal Basics I

Types

Several classifications of signals.

Examples

Some practice problems

Engineers like to classify…

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Why?

It helps identify different classes of problems and develop tools for them…

Let us ask a few questions about signals, each of which will lead to a classification…

1. Is the x-axis continuous or discrete?

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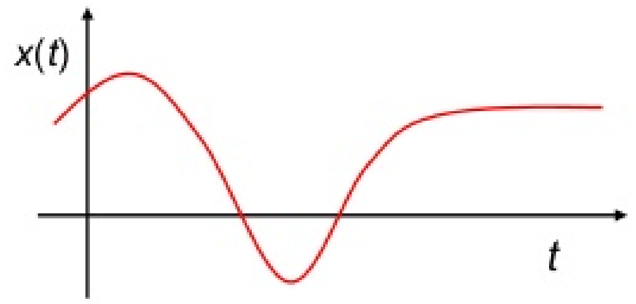
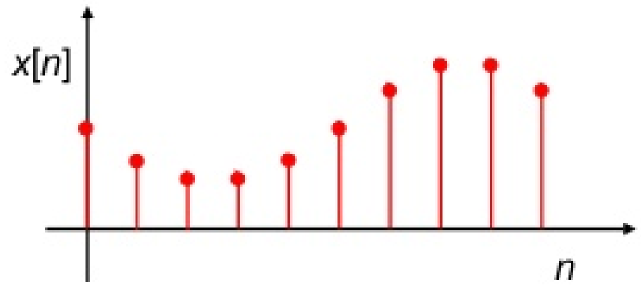


X-axis is a continuous variable (can take any value in a given range).



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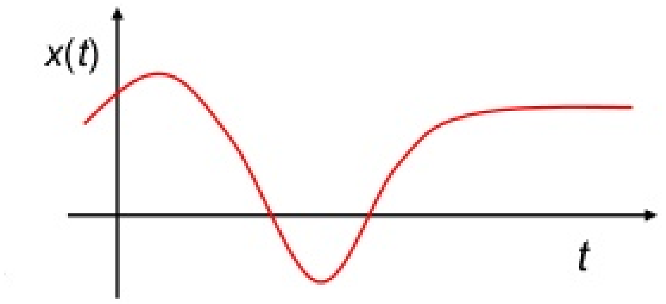
"Continuous-Time"



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Vs.

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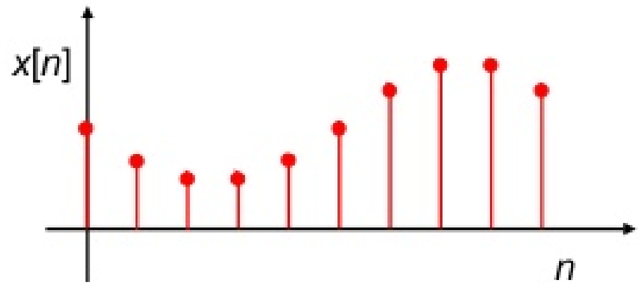

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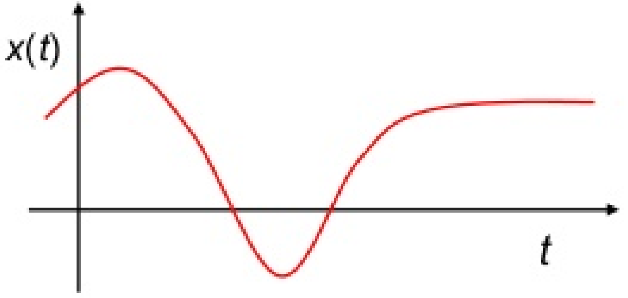

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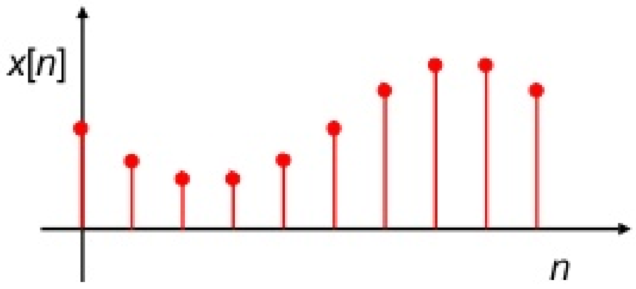

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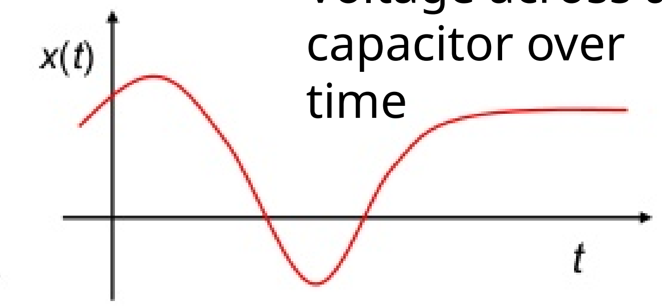
Can you think of an example for each?

1. Is the x-axis continuous or discrete?

“Continuous-Time”

Voltage across a capacitor over time



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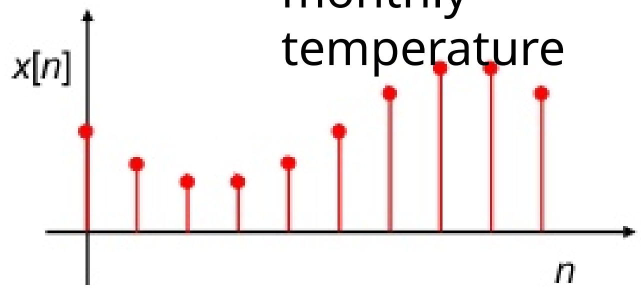
Vs.

“Discrete-Time”

Average monthly temperature



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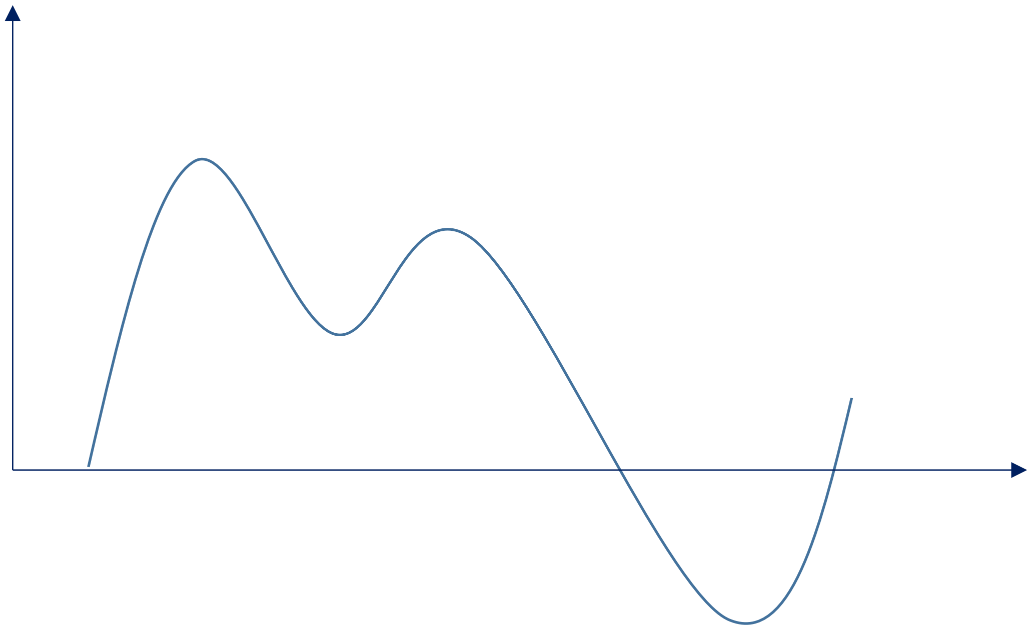

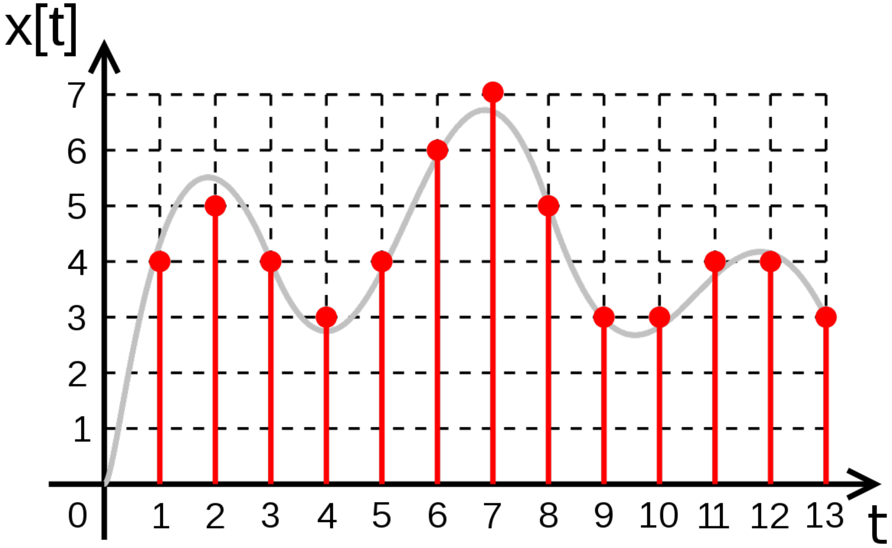

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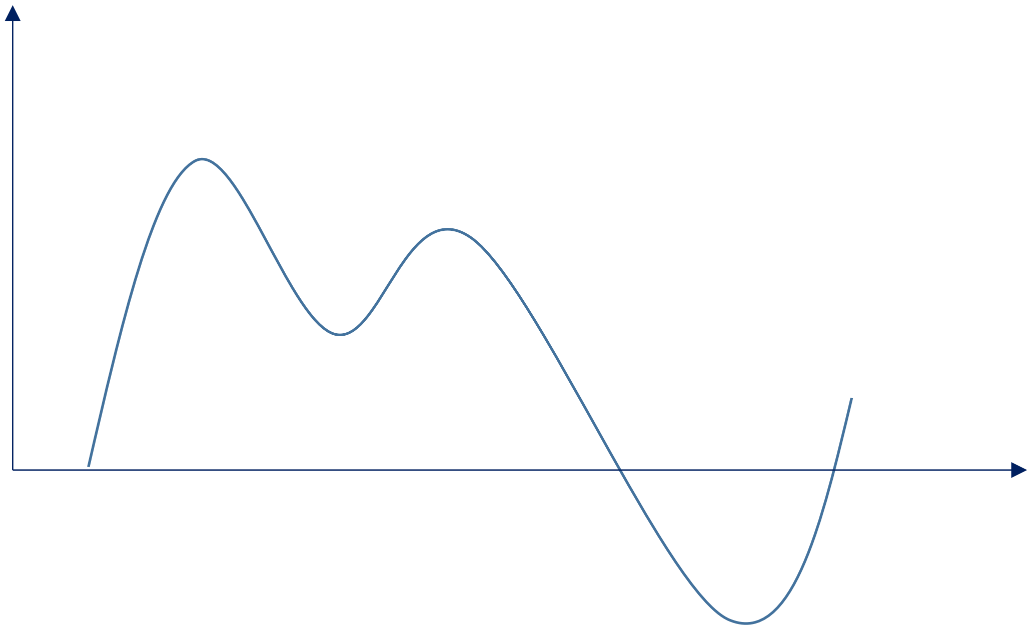
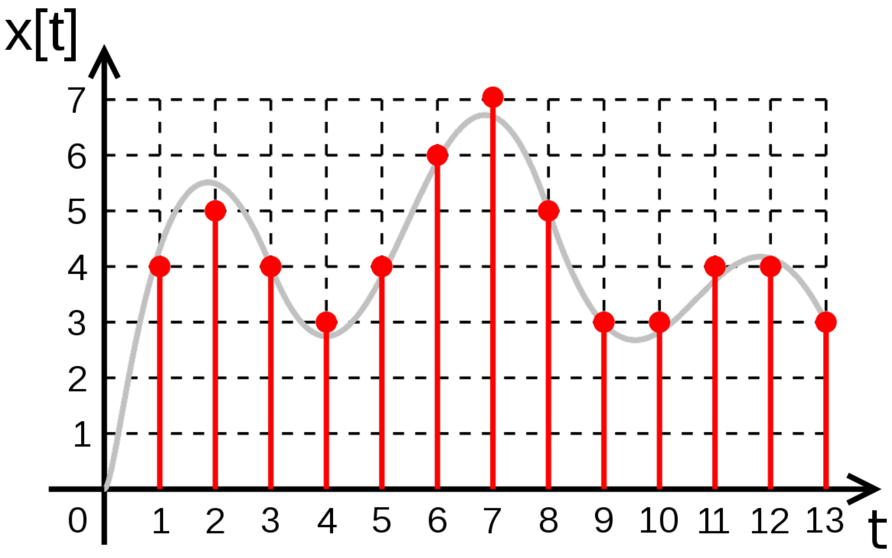
“Analog” Vs. “Digital”





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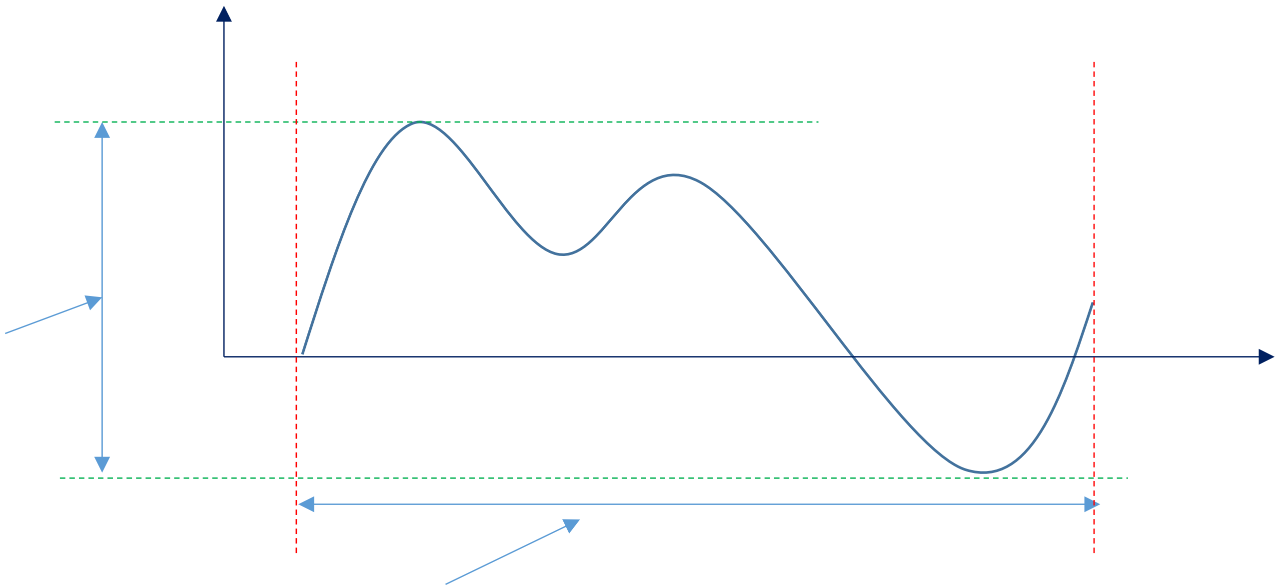
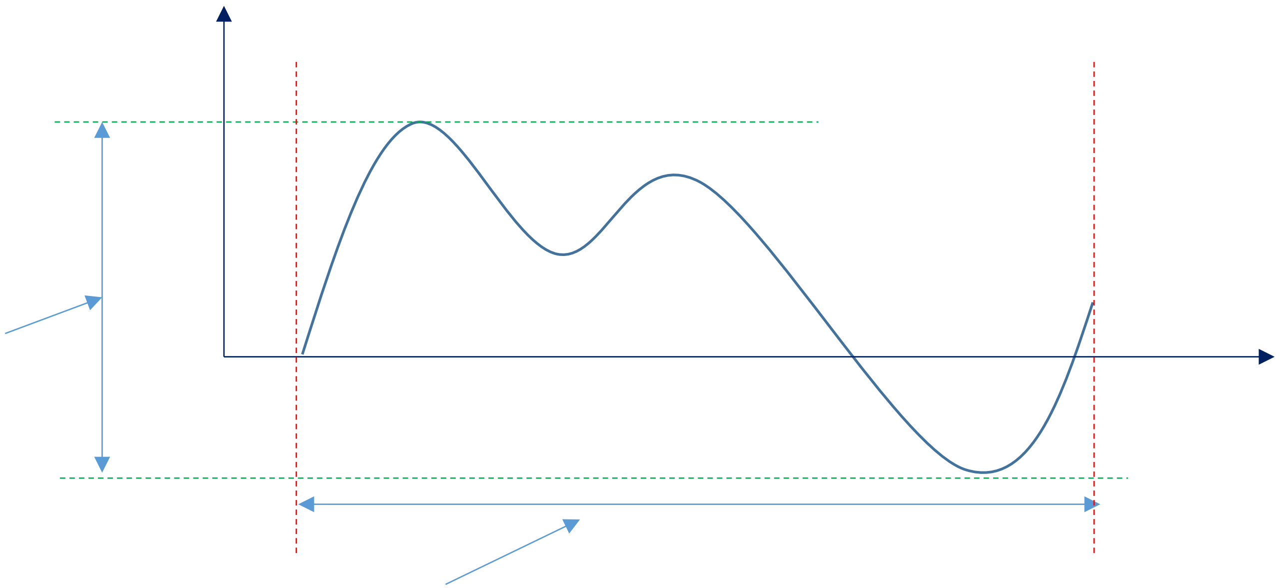


Continuous in time (e.g., can take any value in shown range)

Analog

- Most signals in real life are analog.
  - Analog = both x and y axes are continuous.
  - Continuous? No breaks. Can take **any** value in a given range.
  - e.g., temperature in this room

Continuous in amplitude (e.g., can take any value in shown range)



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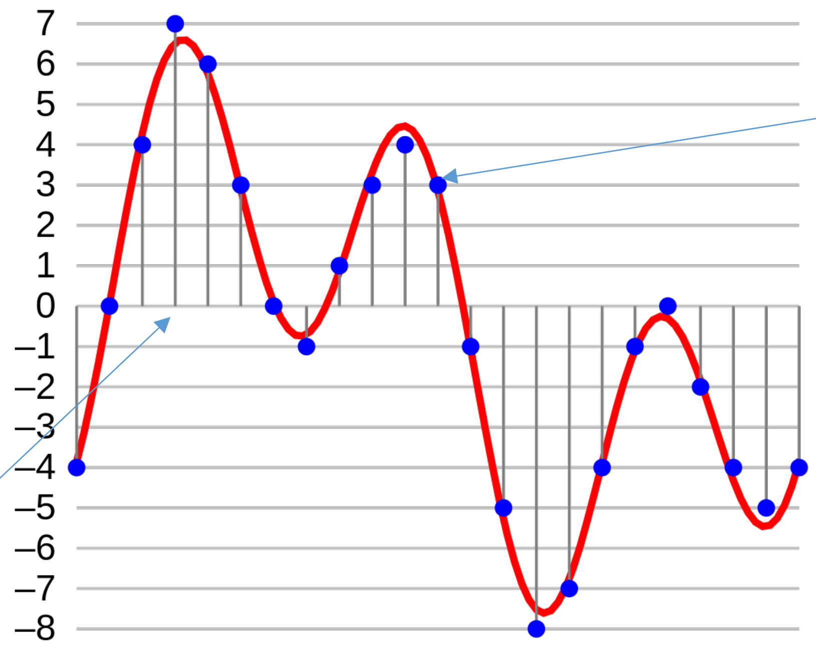

“discontinuous” in amplitude (e.g., can not take values between 3 and 4, like 3.5 etc.)

discontinuous in time (e.g., values taken only at fixed times but not in-between)

Digital Logic in an Analog World

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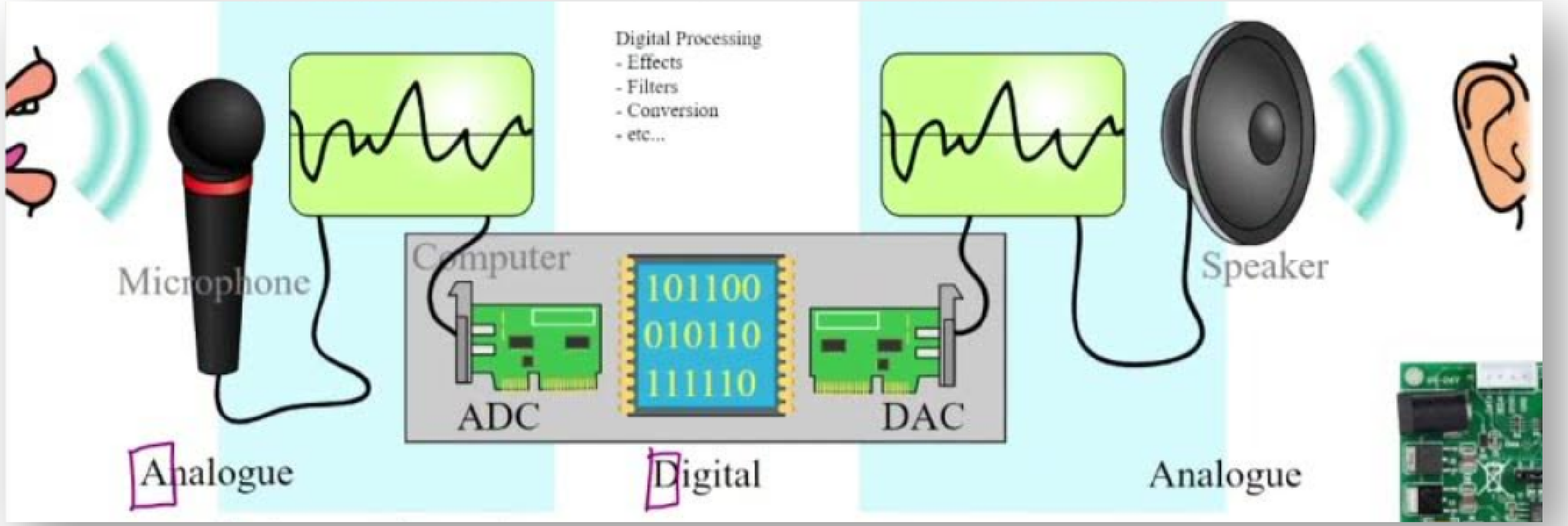

https://www.youtube.com/@ETphysics

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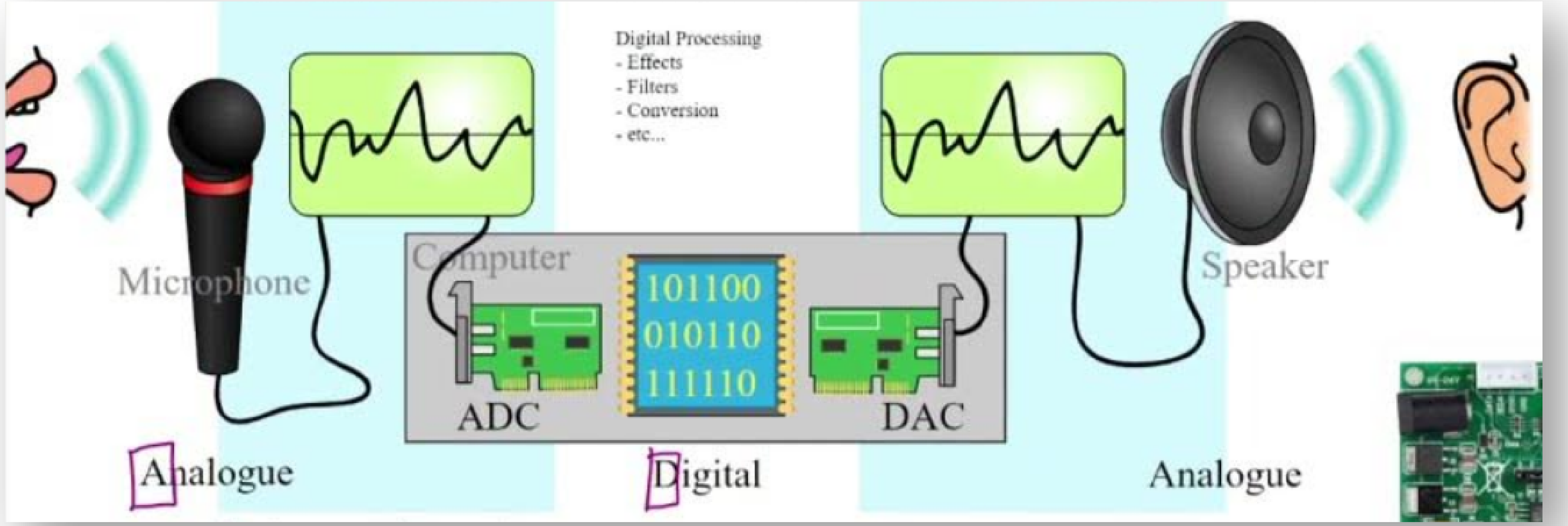

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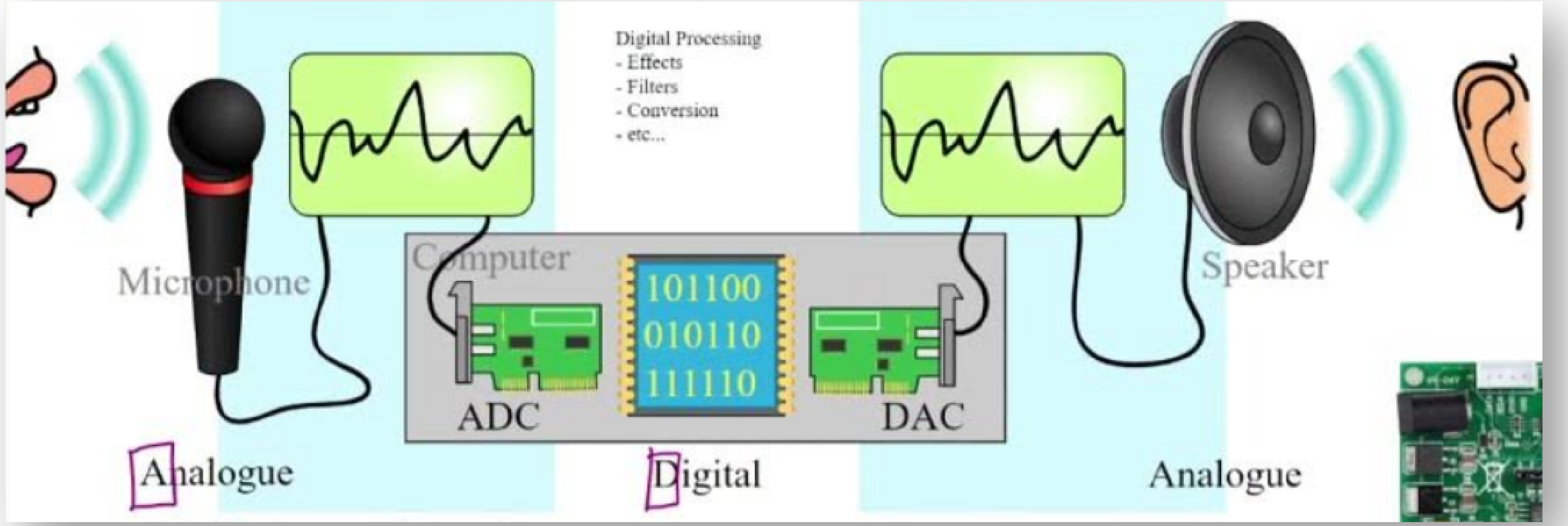

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- But mostly: we digitize data by sampling/discretizing continuous signals/systems.

3.Does the signal have a repeating pattern?

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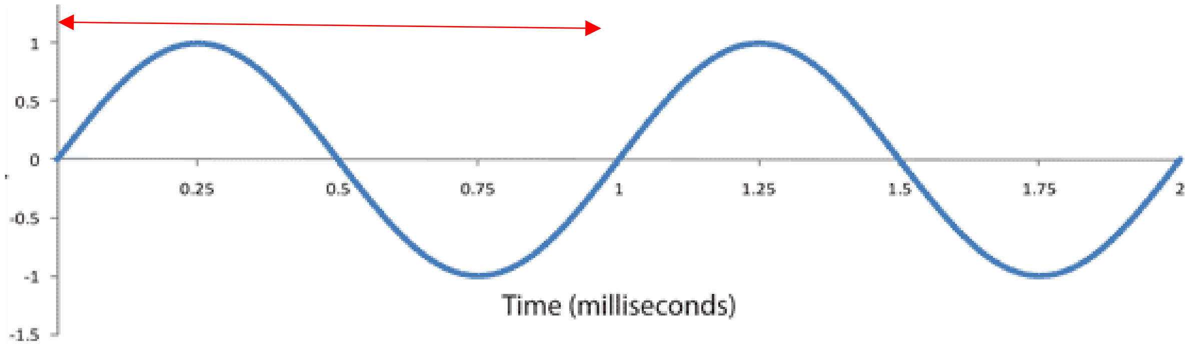
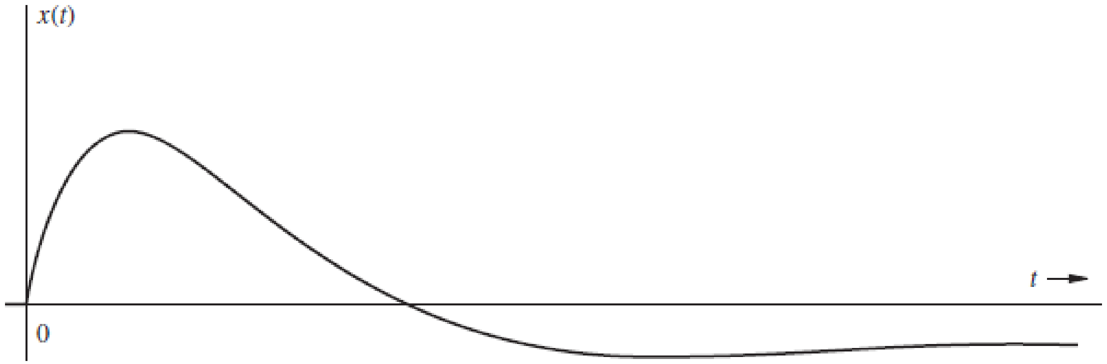
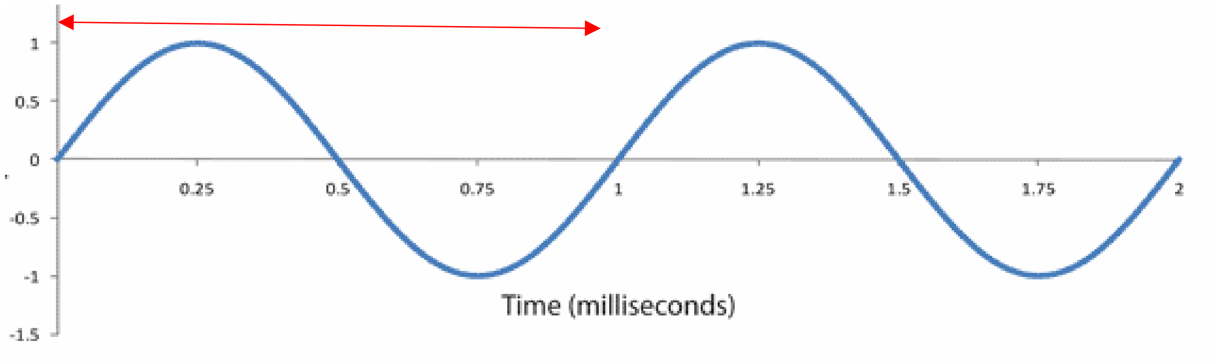




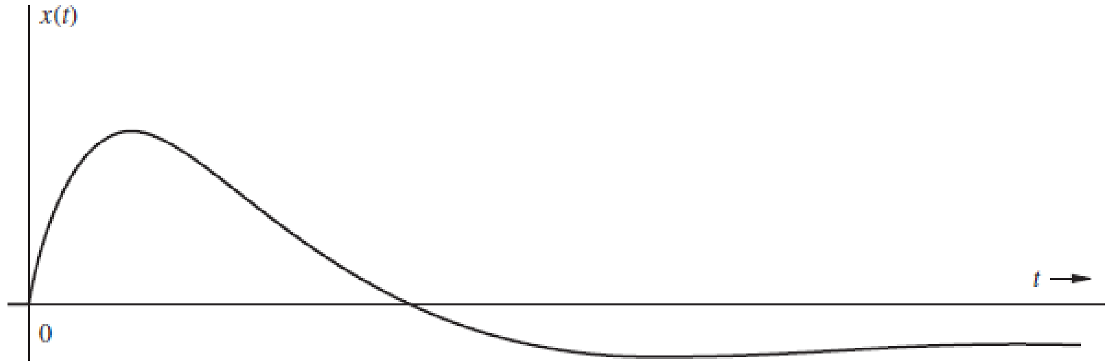
3.Does the signal have a repeating pattern?

"Periodic"



Vs.

"Aperiodic"



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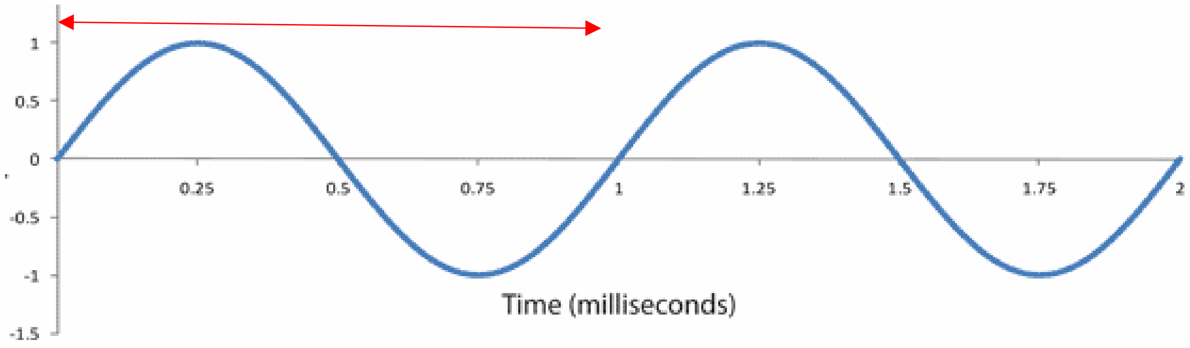

Vs.

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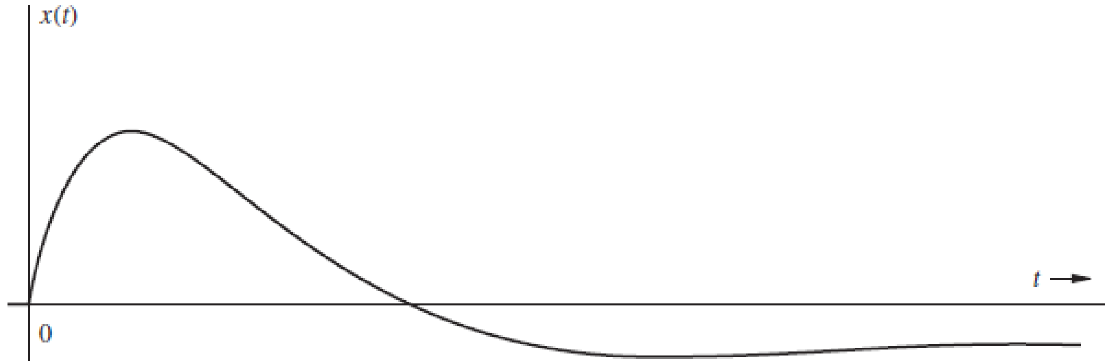
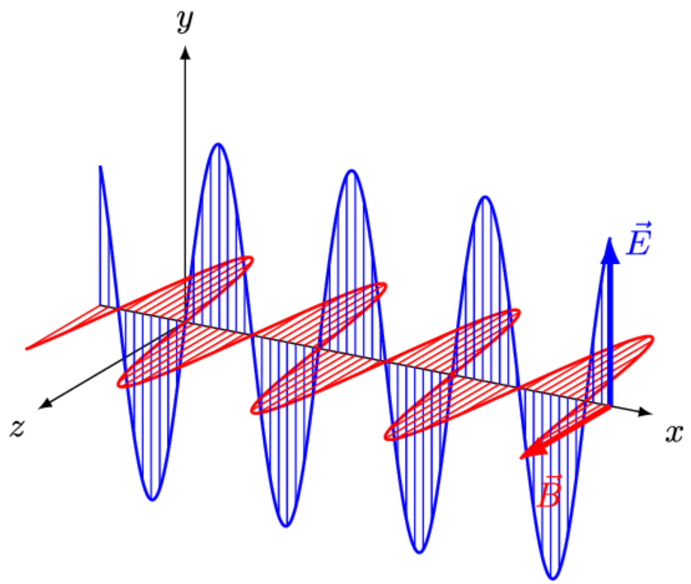


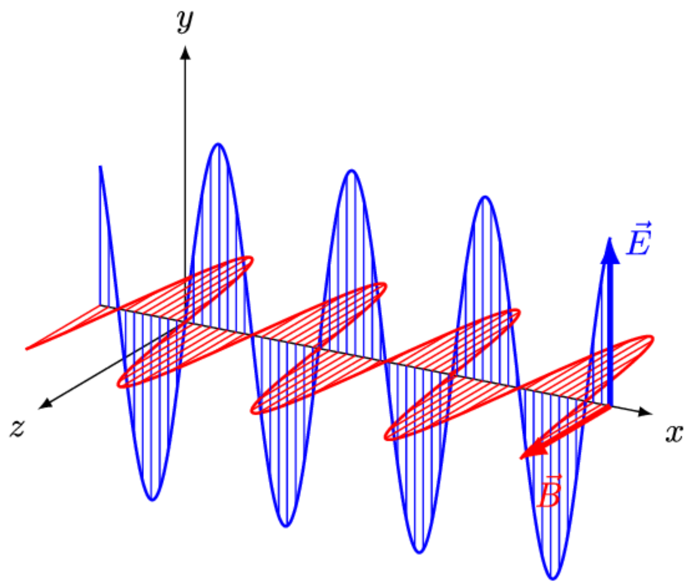
Can you think of a periodic and an aperiodic signal?

Periodic



EM Waves

Periodic



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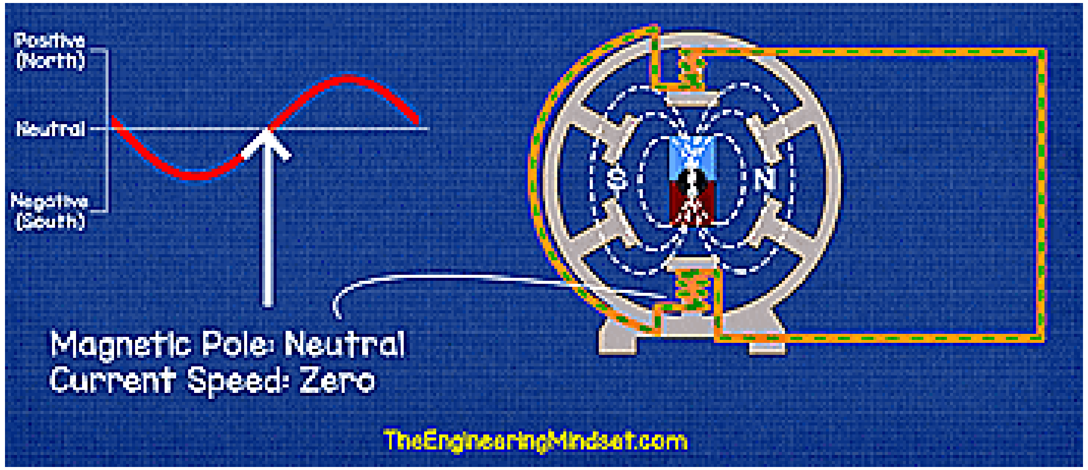
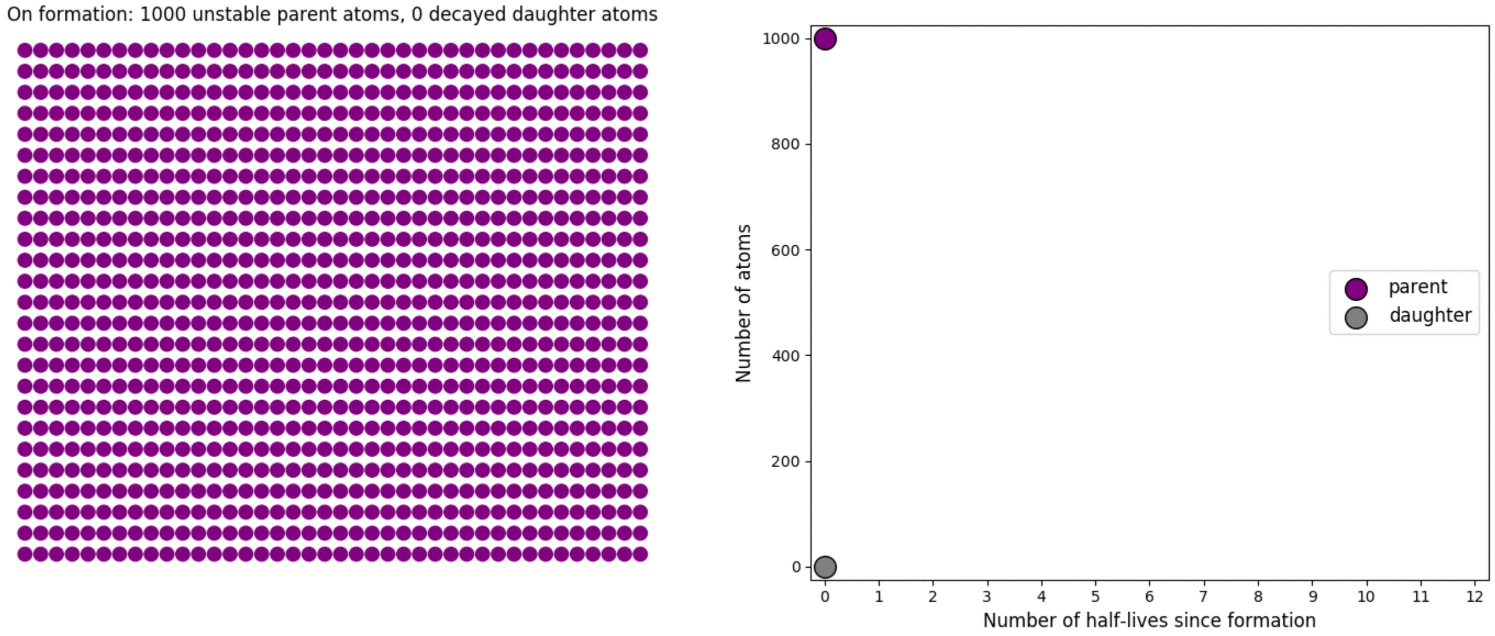
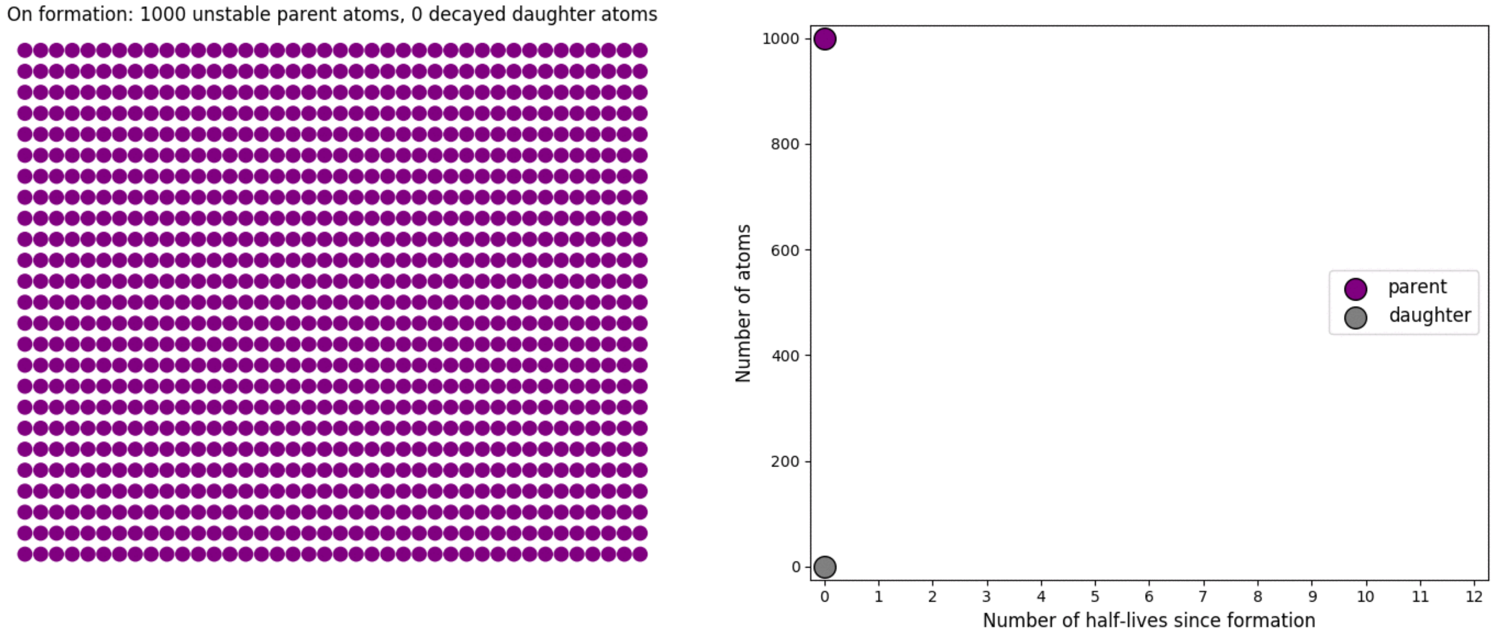


Alternating Current



Radioactive Decay

Aperiodic



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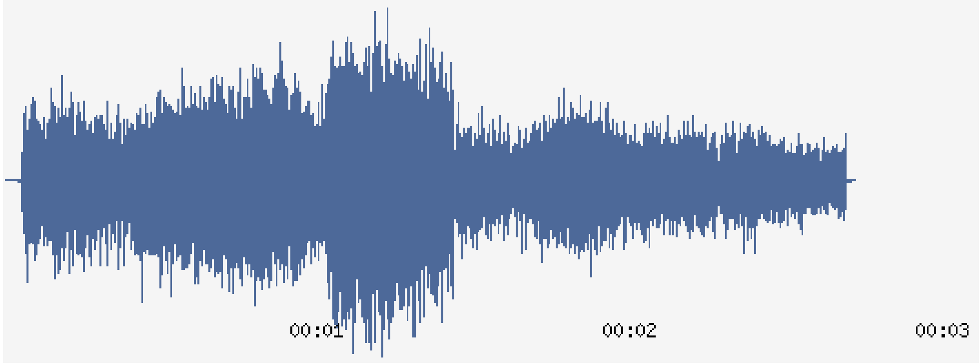

Car Crash Recording

Periodic Signal's Time Period & Frequency

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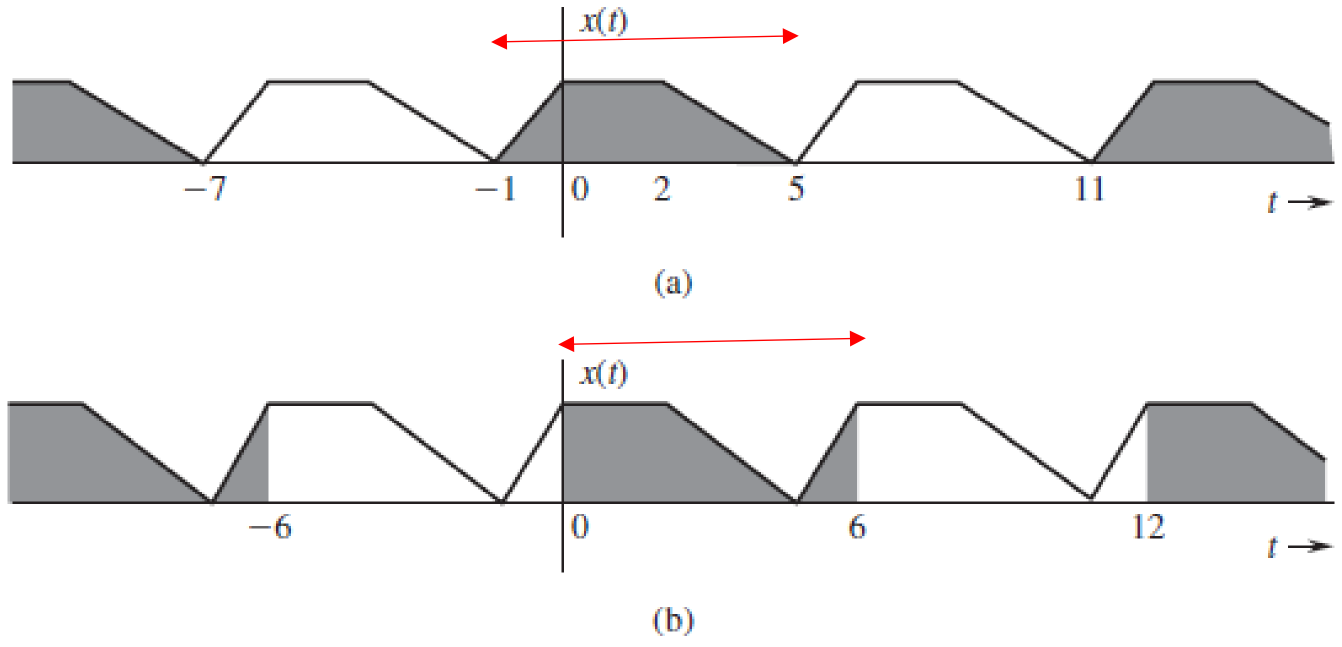

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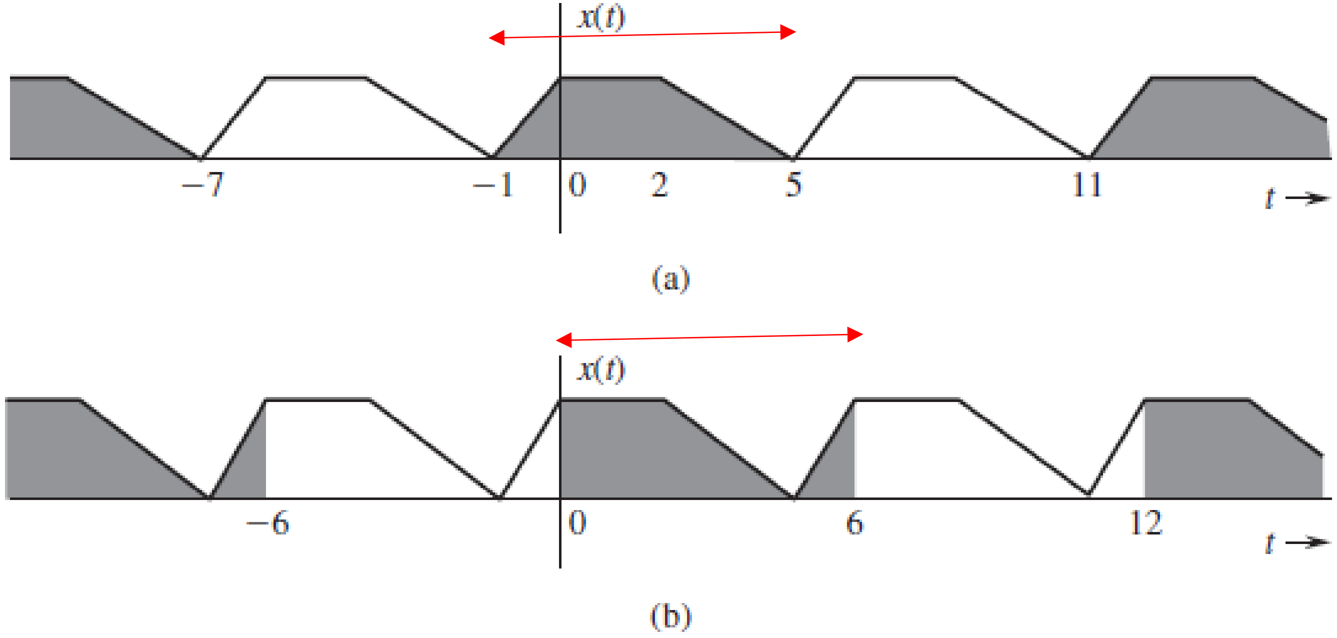

$T_0 =$ time period = length of the minimum repeating pattern.

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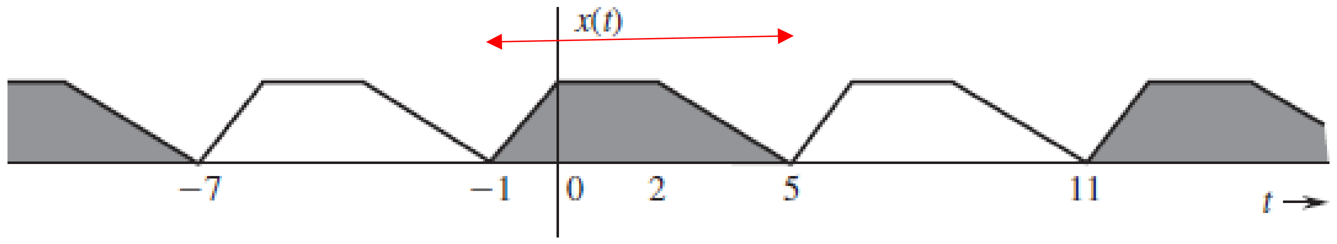
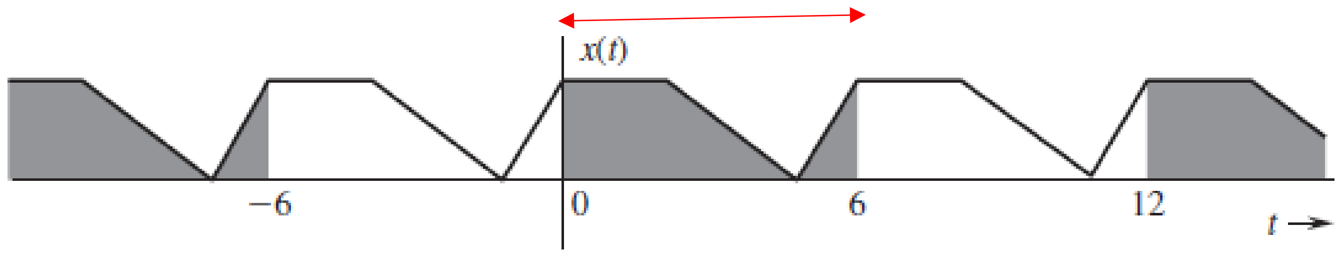
Periodic Signal's Time Period & Frequency



(a)



(b)

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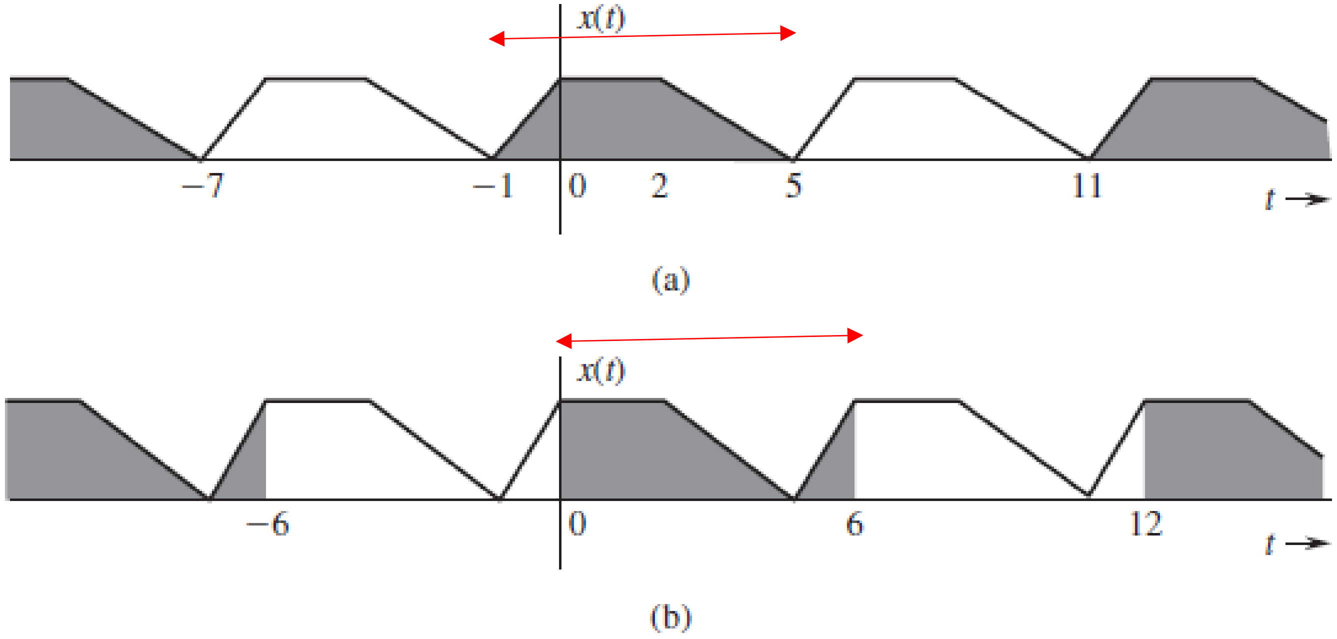
Can you write a mathematical definition of periodicity?

Periodic Signal's Time Period & Frequency



T_0 = time period = length of the minimum repeating pattern.

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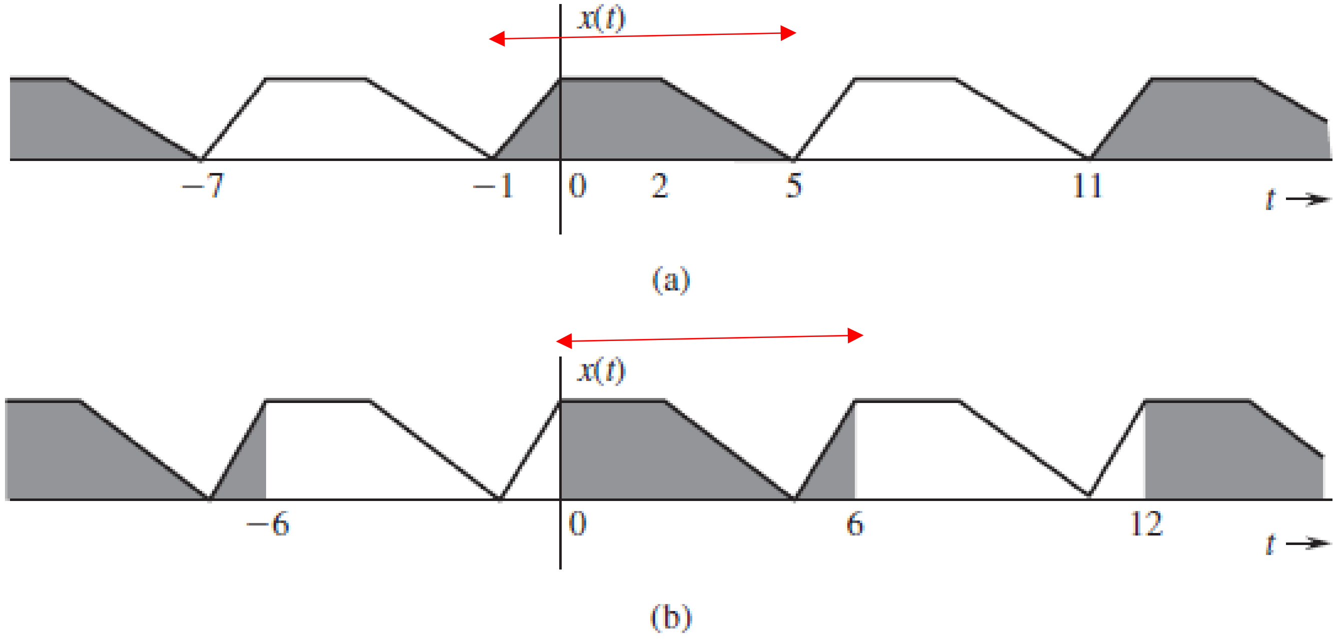
$$\boxed{x(t) = x(t + T_0) \qquad \text{for all } t}$$

Periodic Signal's Time Period & Frequency



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Mathematical requirement for a function $x(t)$ to be periodic.

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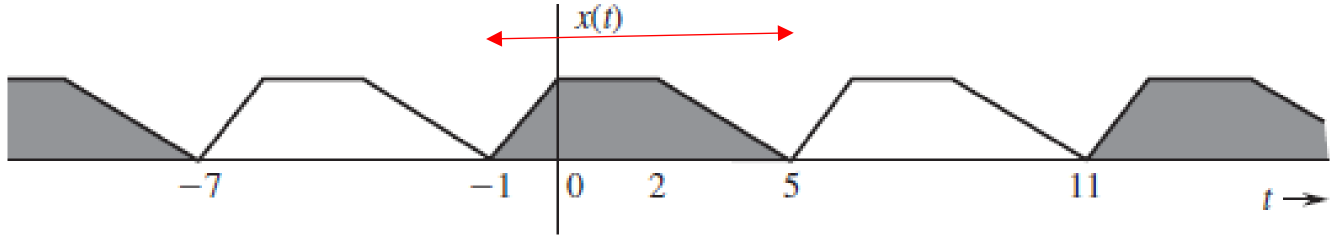

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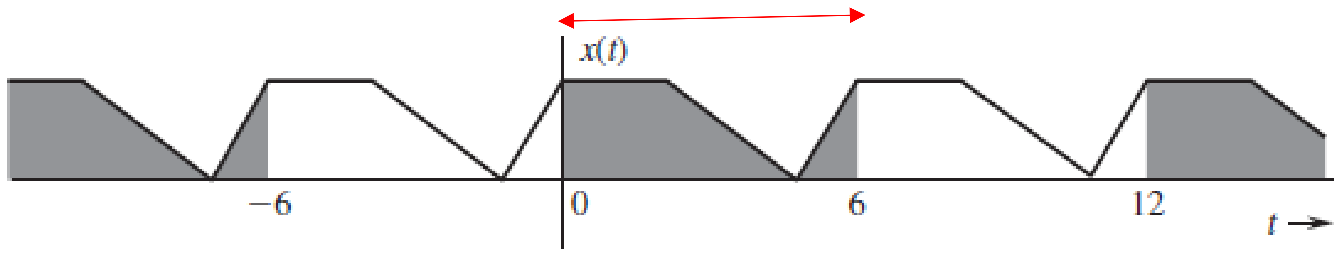

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Keep in mind that this should hold for all t!

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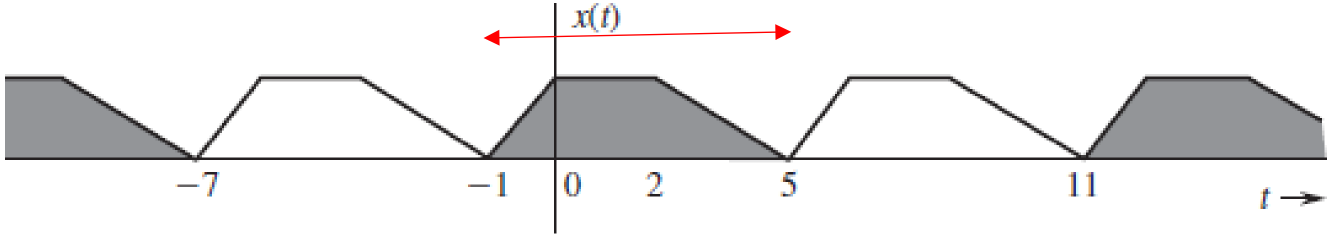

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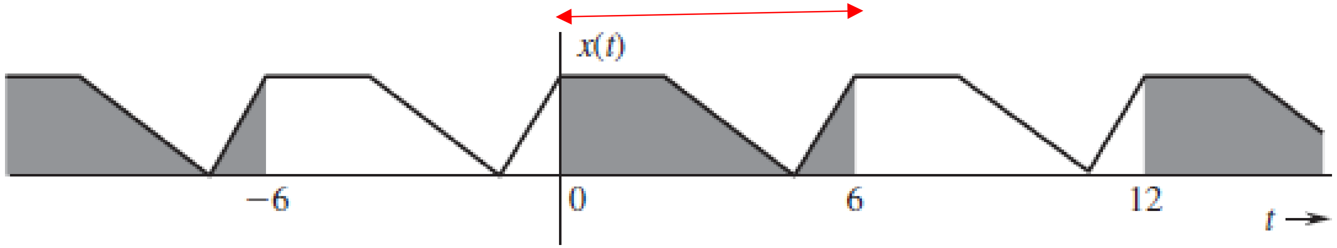

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Keep in mind that this should hold for all t!

e.g., for signal to be periodic with period $T_0 = 6$, we must have

$$x(0) = x(0 + 6)$$
$$x(1) = x(1 + 6)$$
$$x(-1) = x(-1 + 6)$$
$$\ldots$$

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Therefore, $\cos(t)$ is periodic with period

$$T_0 = 2\pi$$

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$$\boxed{RHS = \cos\left(\omega_0\left(t + \frac{2\pi}{\omega_0}\right)\right) = \cos(\omega_0 t + 2\pi) = \cos(\omega_0 t) = LHS}$$

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Q.E.D

ω_0 vs. f_0

ω_0 $vs.$ f_0

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- ω_0 is called "angular frequency" and is given in radians/second.

Visualizing relation between t and θ

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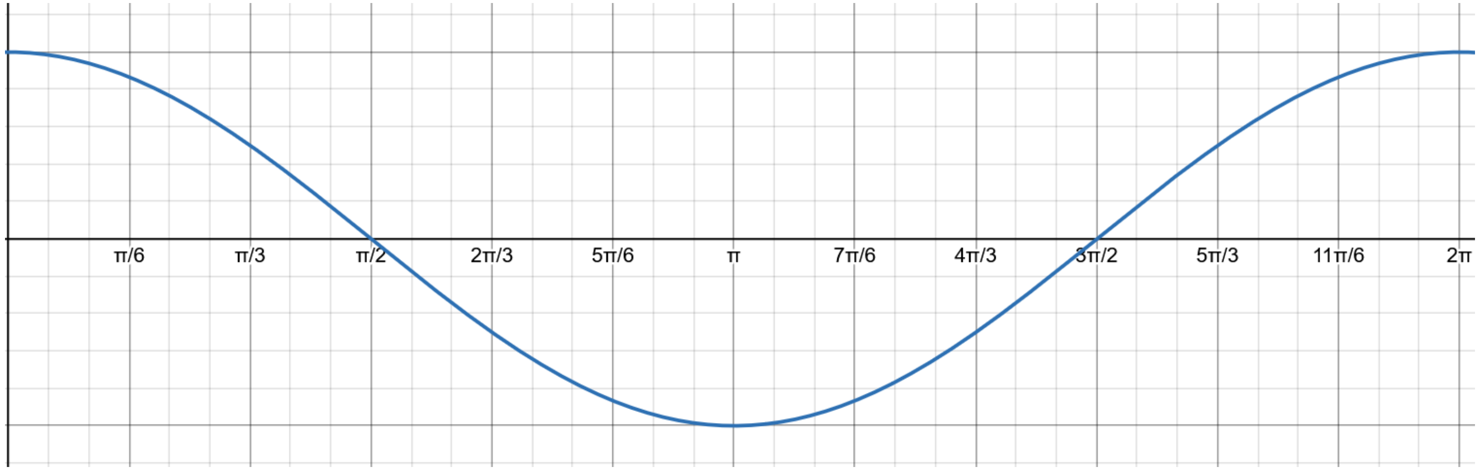

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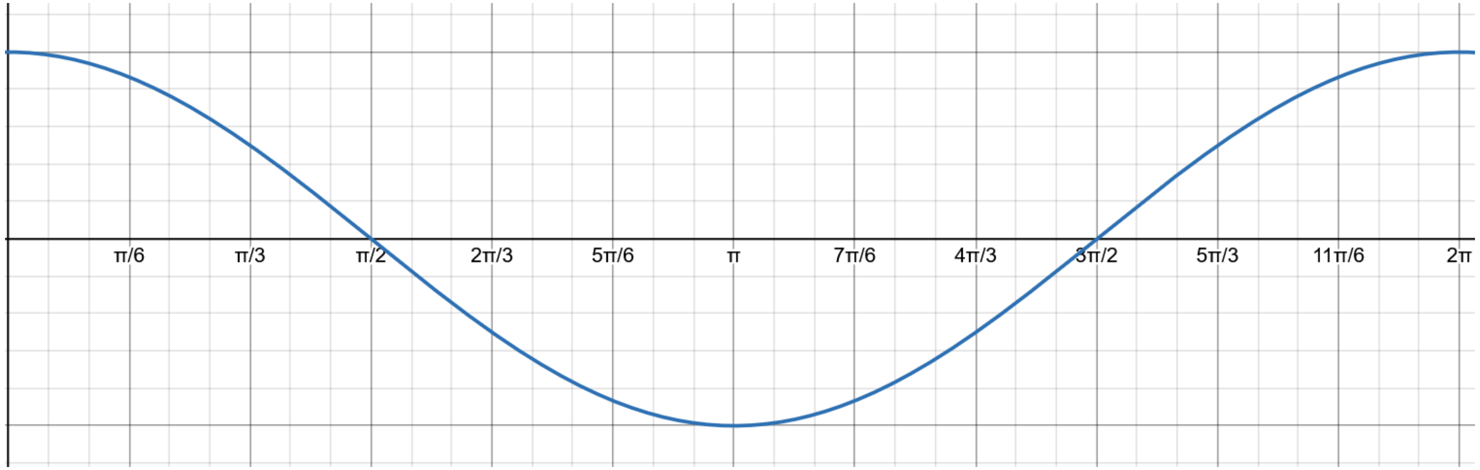

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As time goes from 0 to 2π, the angle θ also goes from 0 to 2π, and $\cos(t)$ completes one cycle.

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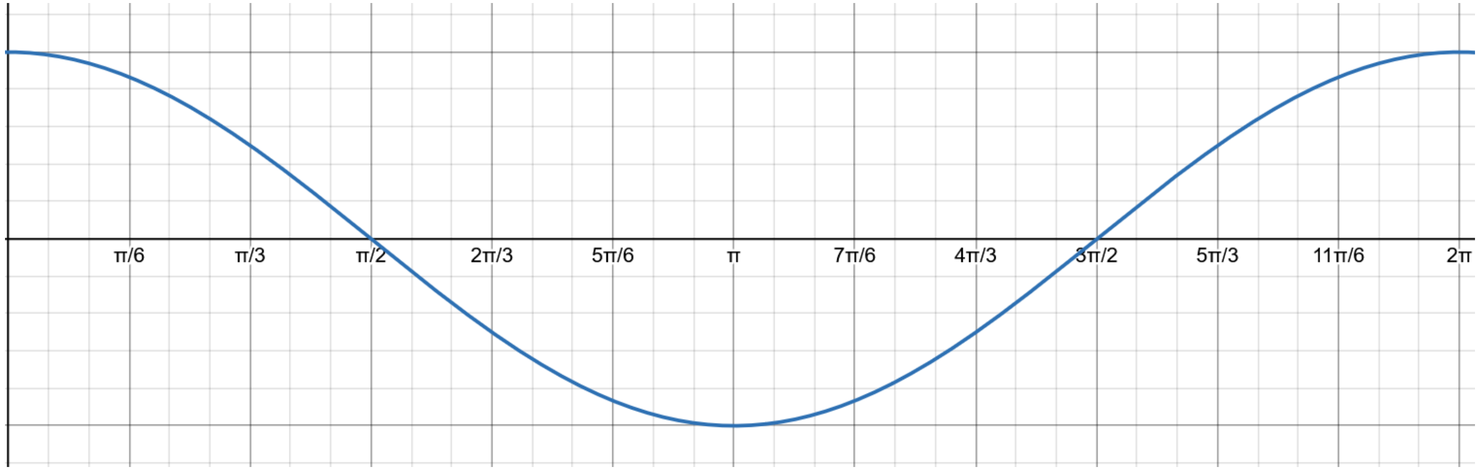

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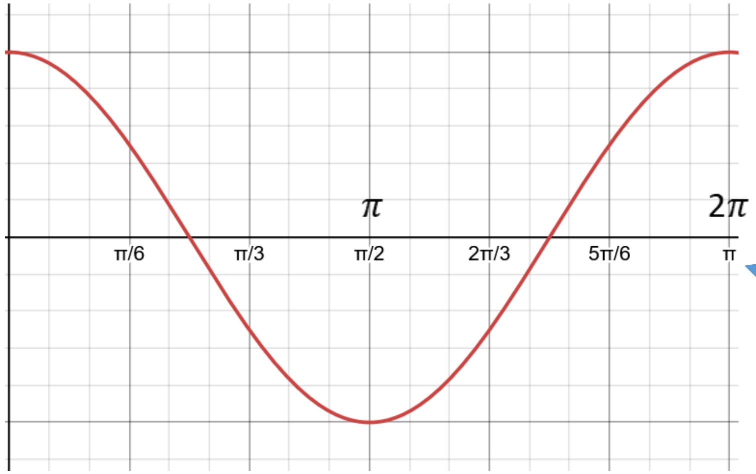

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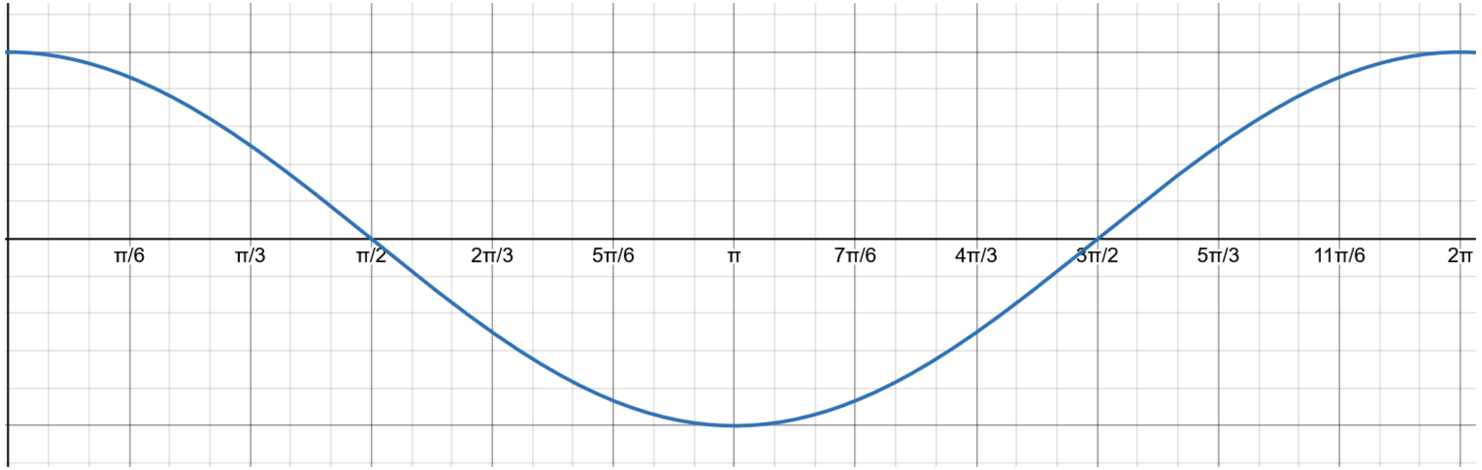

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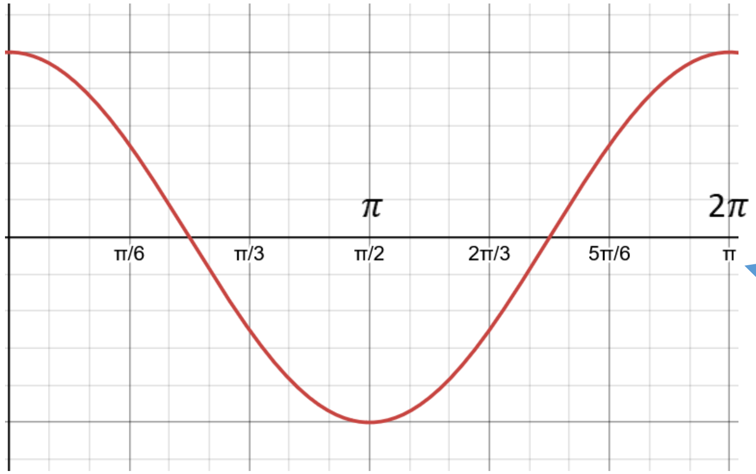

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As time goes from 0 to π, the angle θ goes from 0 to 2π, thus $\cos(2t)$ completes one cycle much faster.

Q. Is the sum of periodic signals always periodic?

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Example 1

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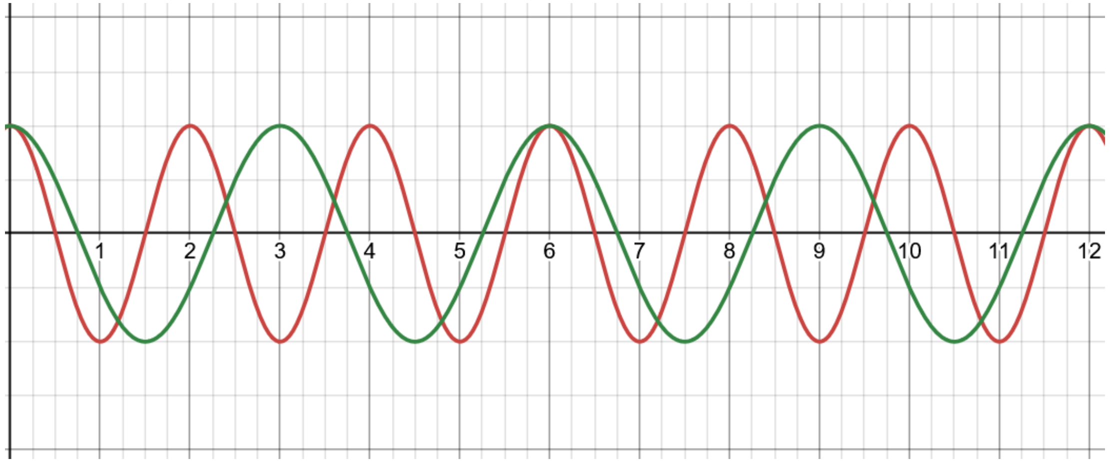

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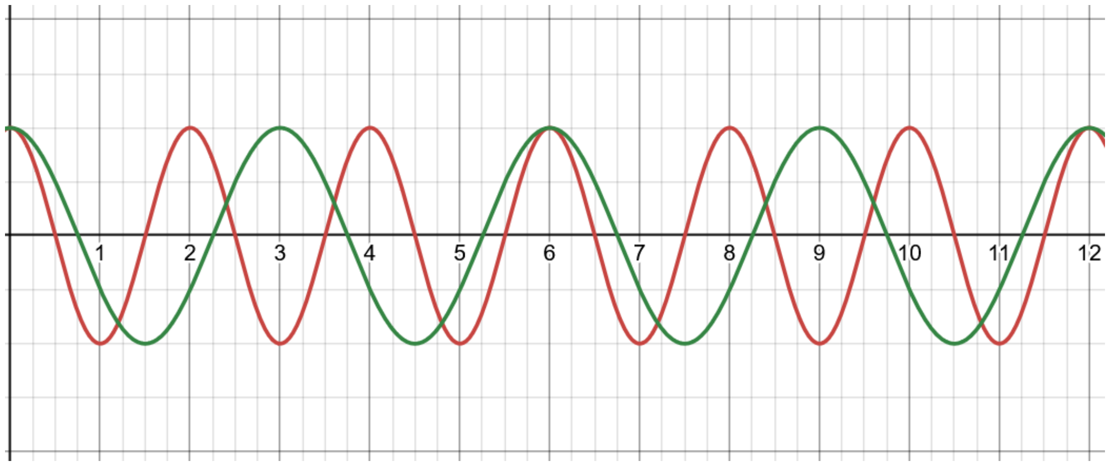

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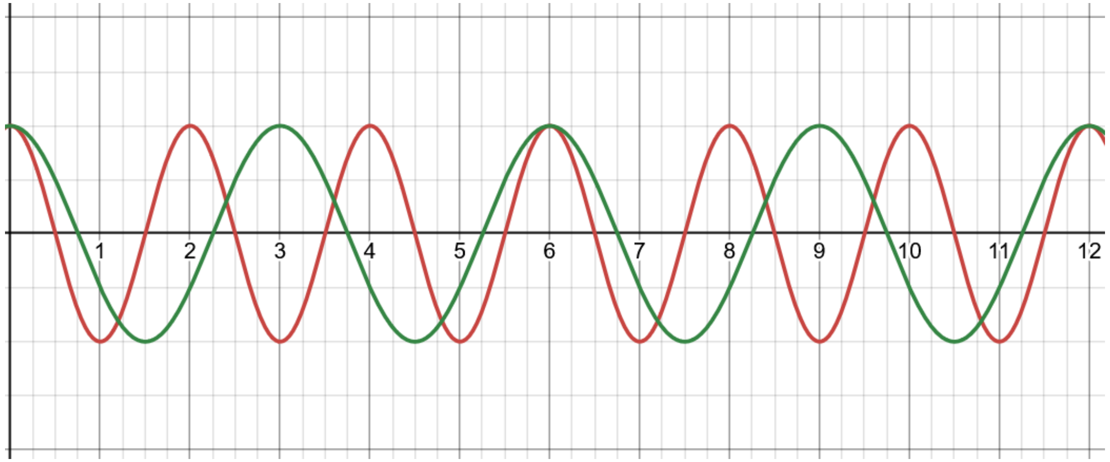

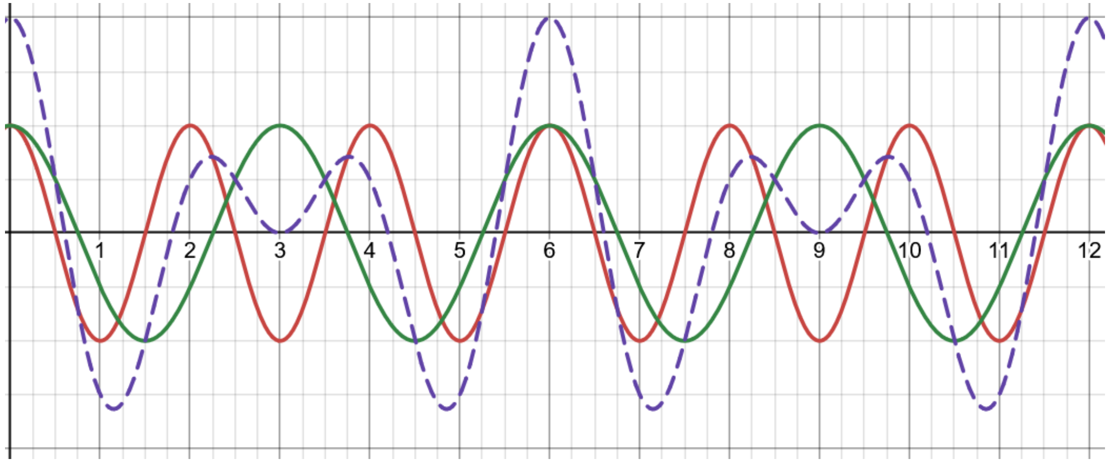

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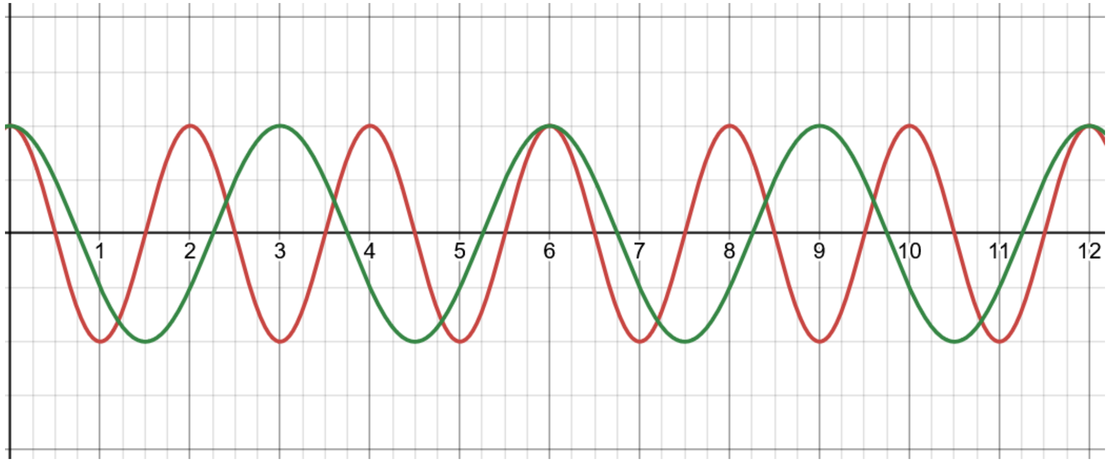

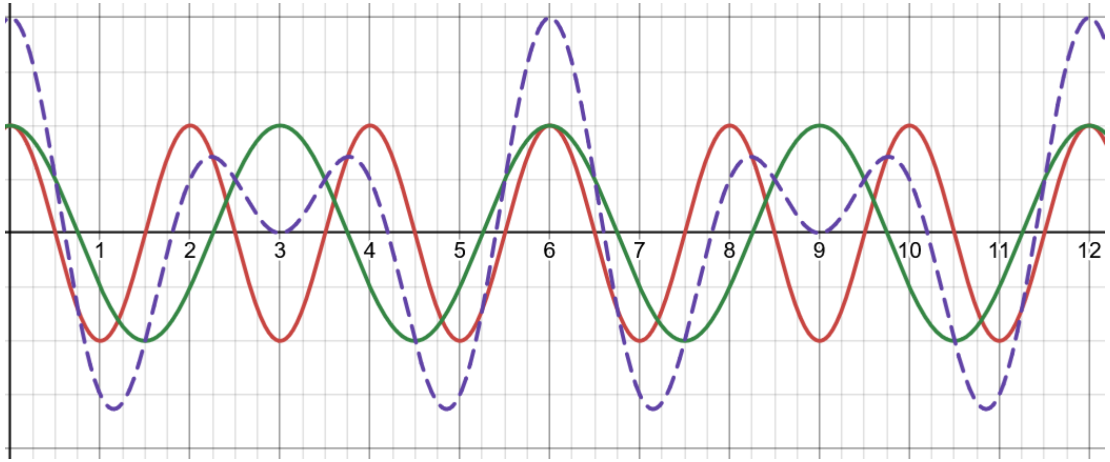

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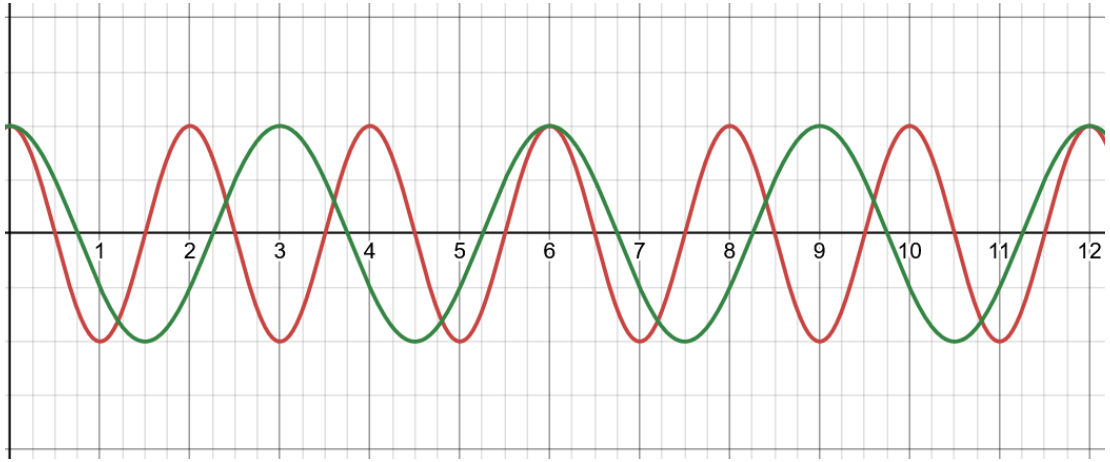

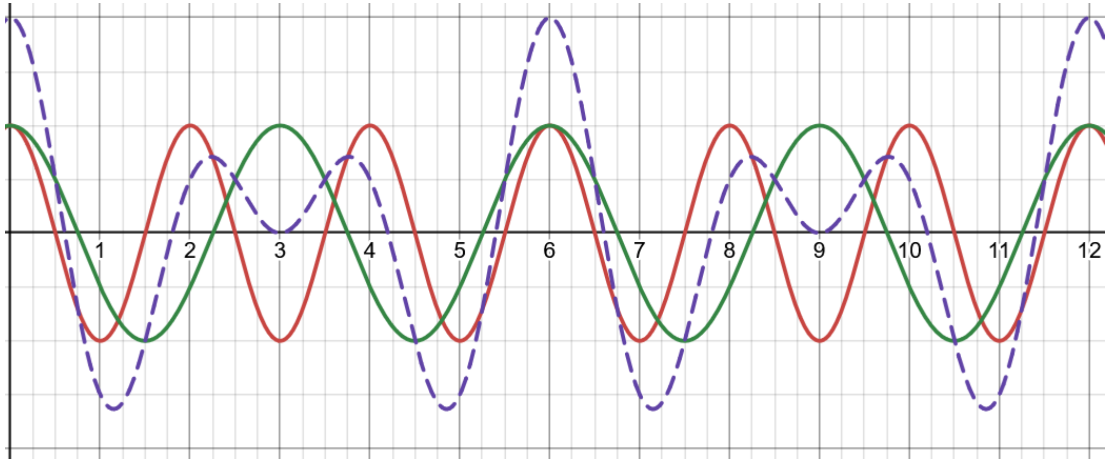

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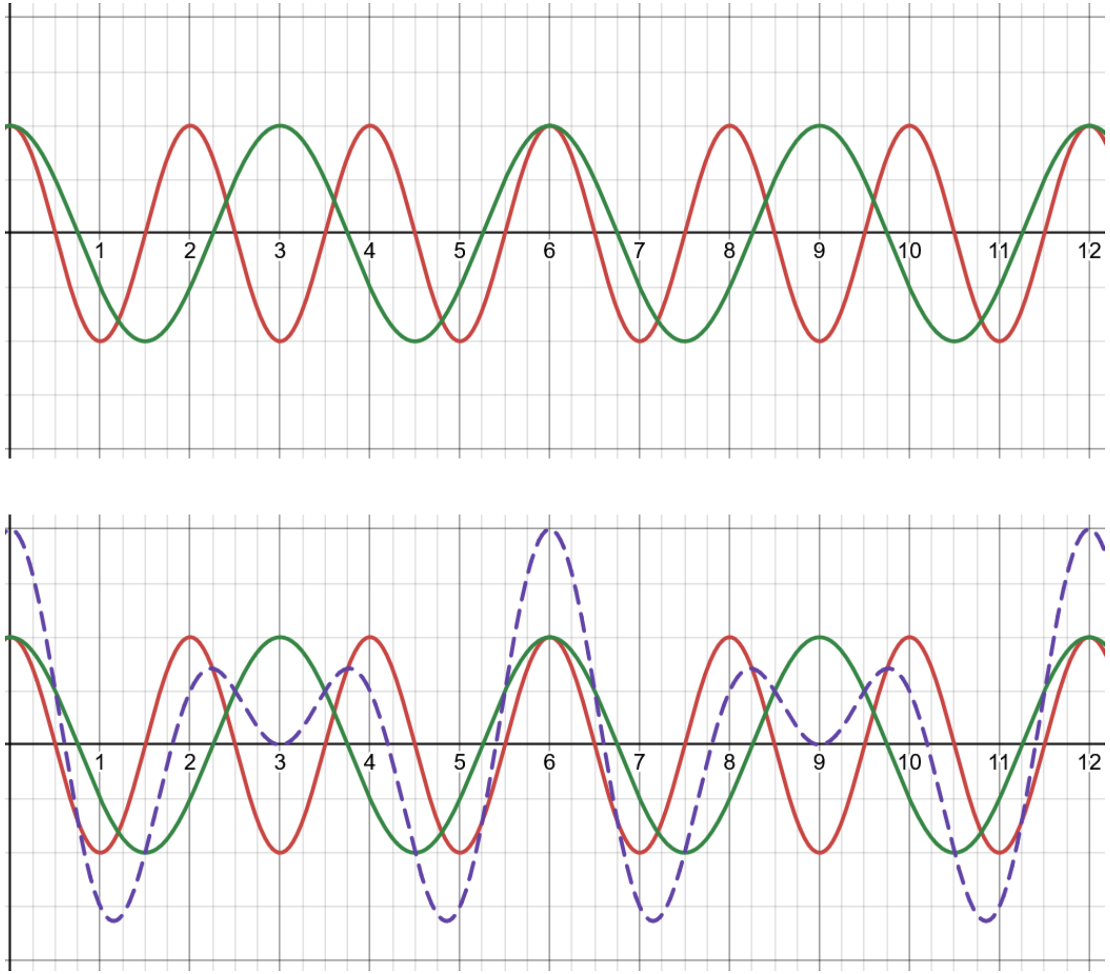

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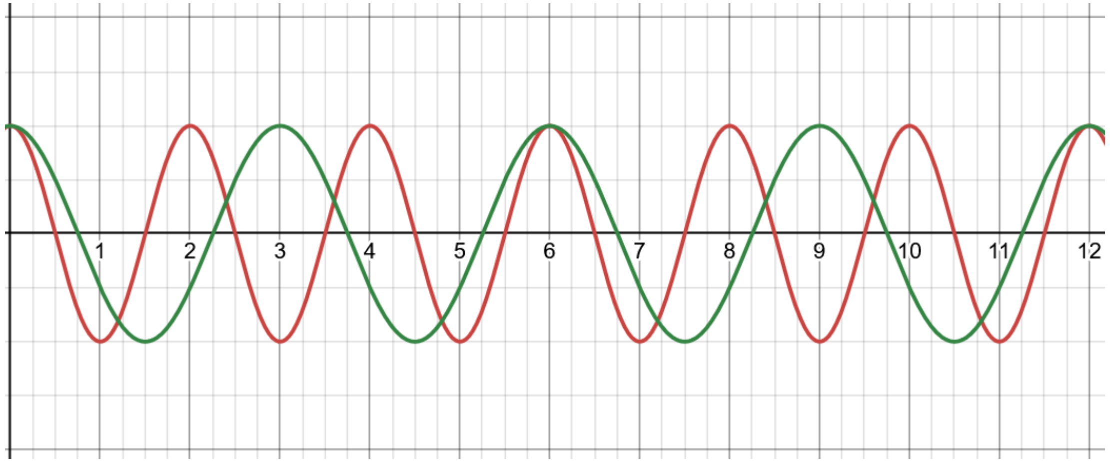

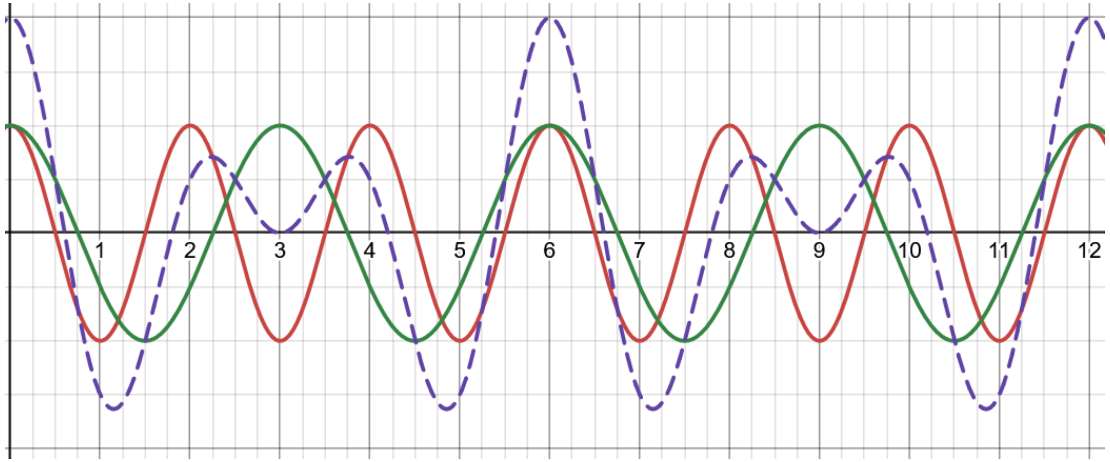

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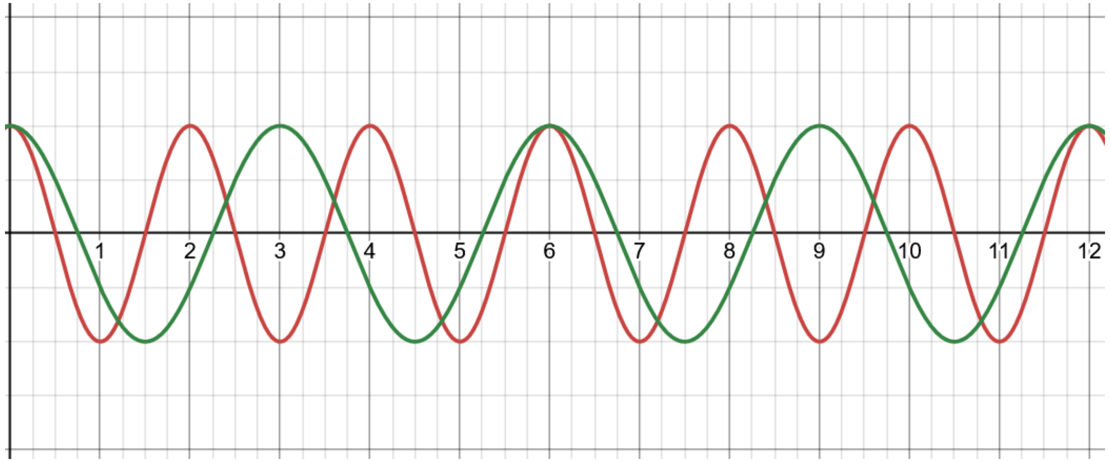

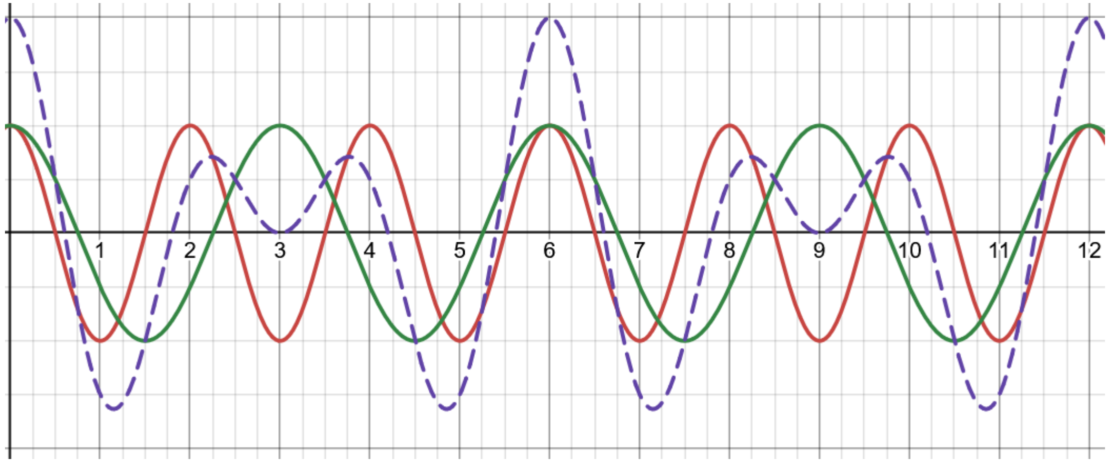

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At 6, their cycles match (one has completed three cycles and other has completed two)

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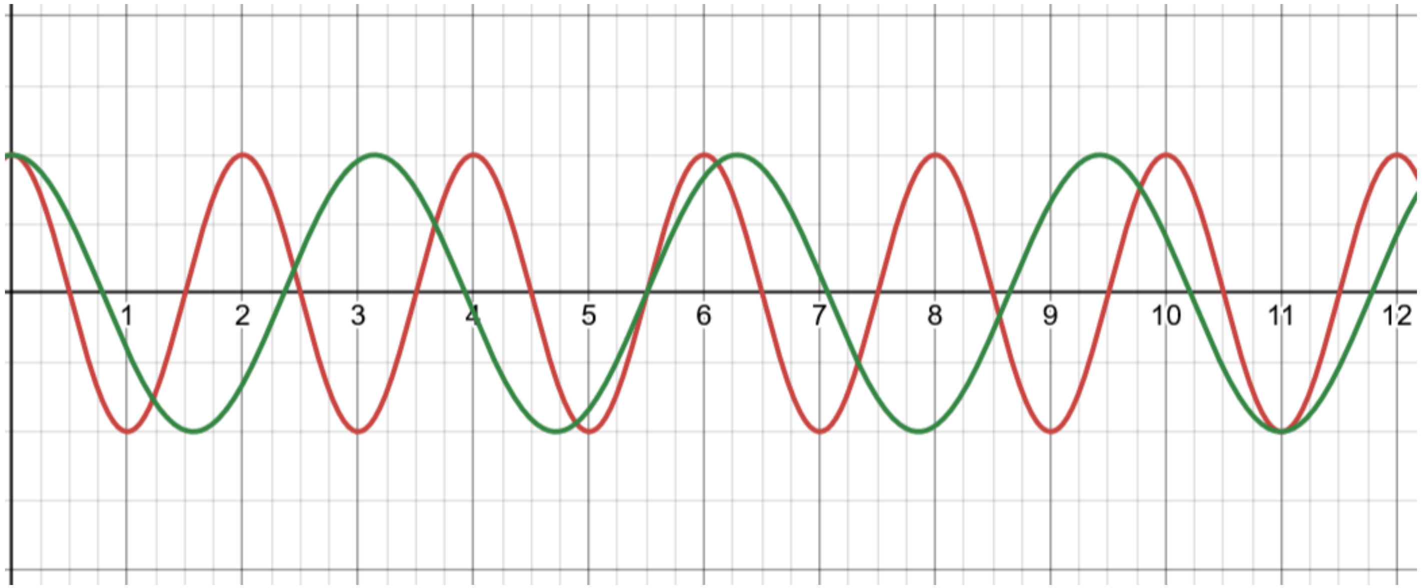

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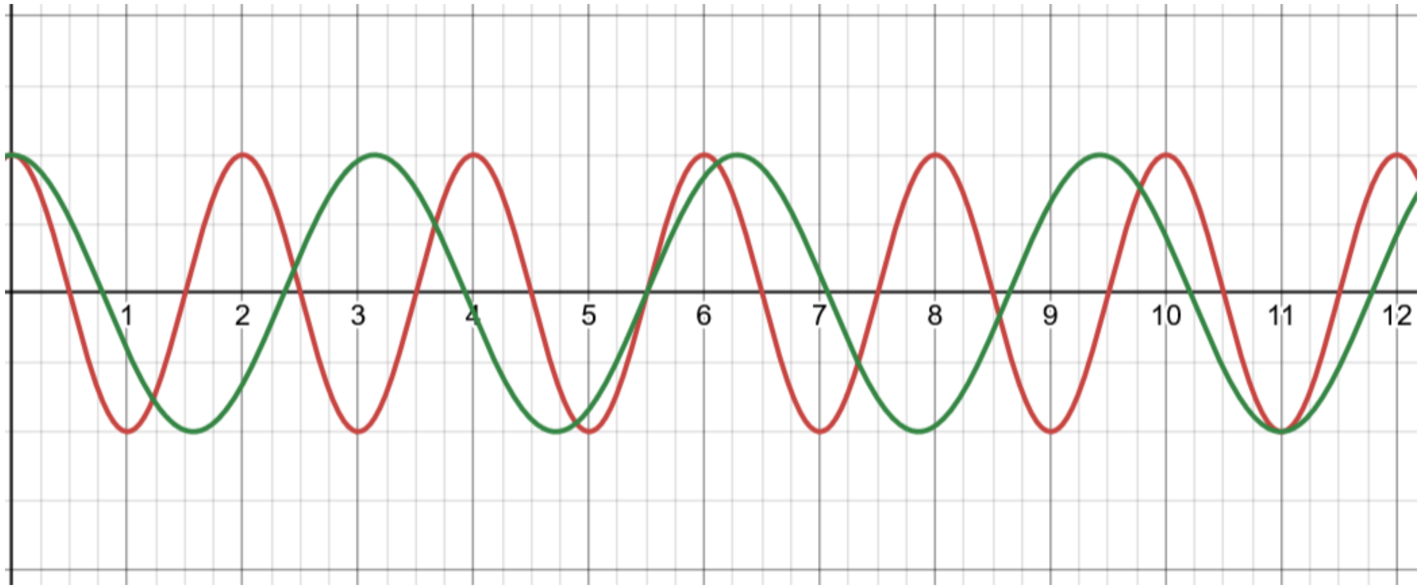

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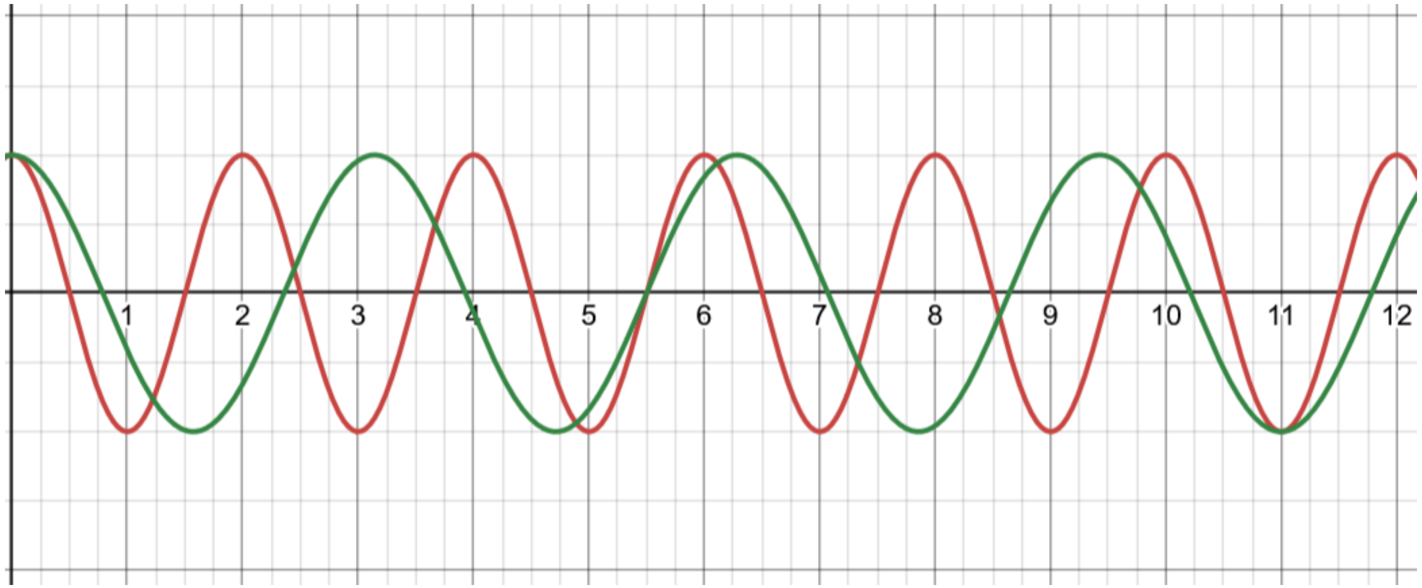

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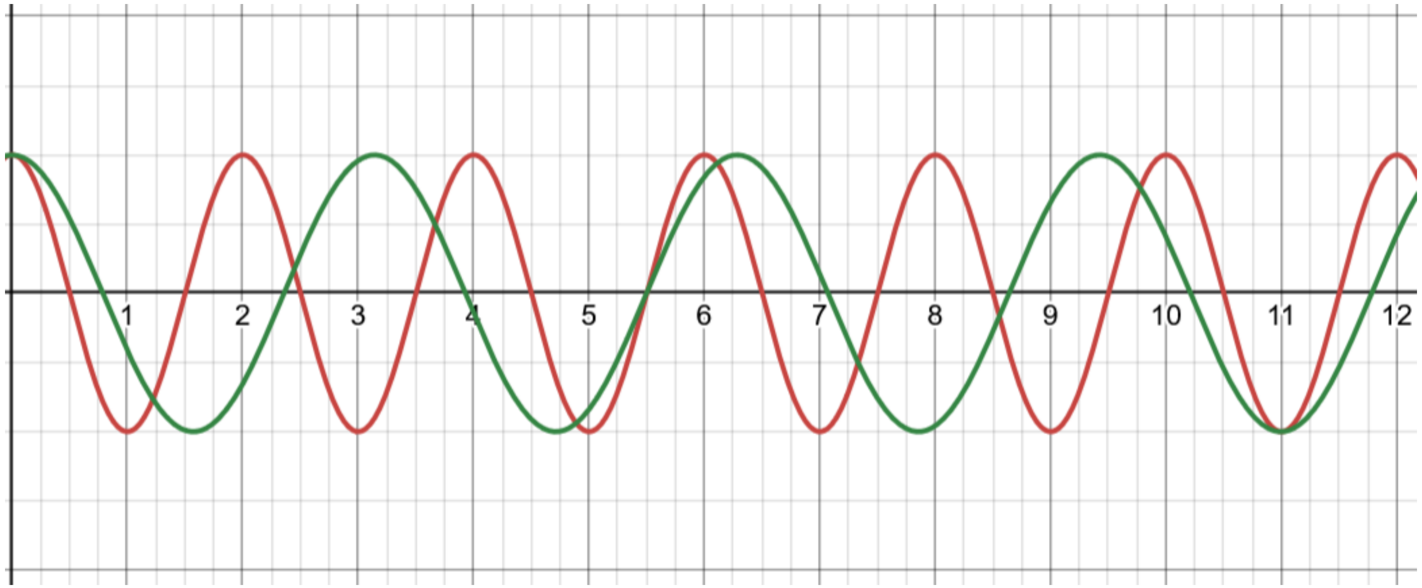

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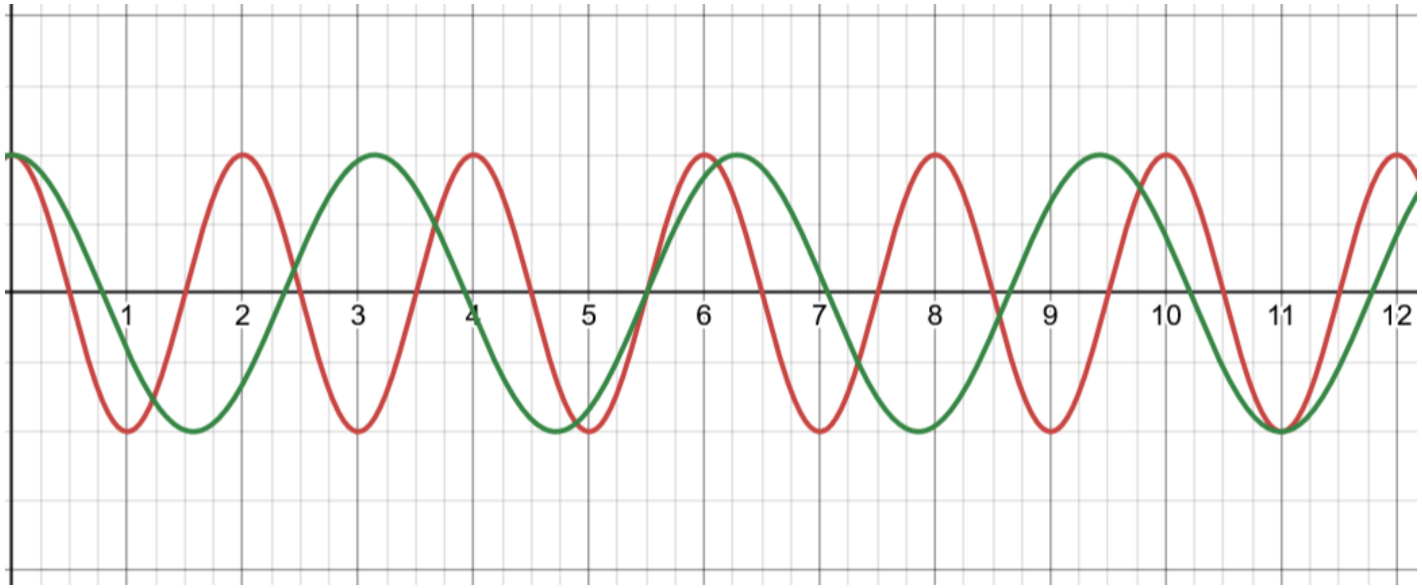

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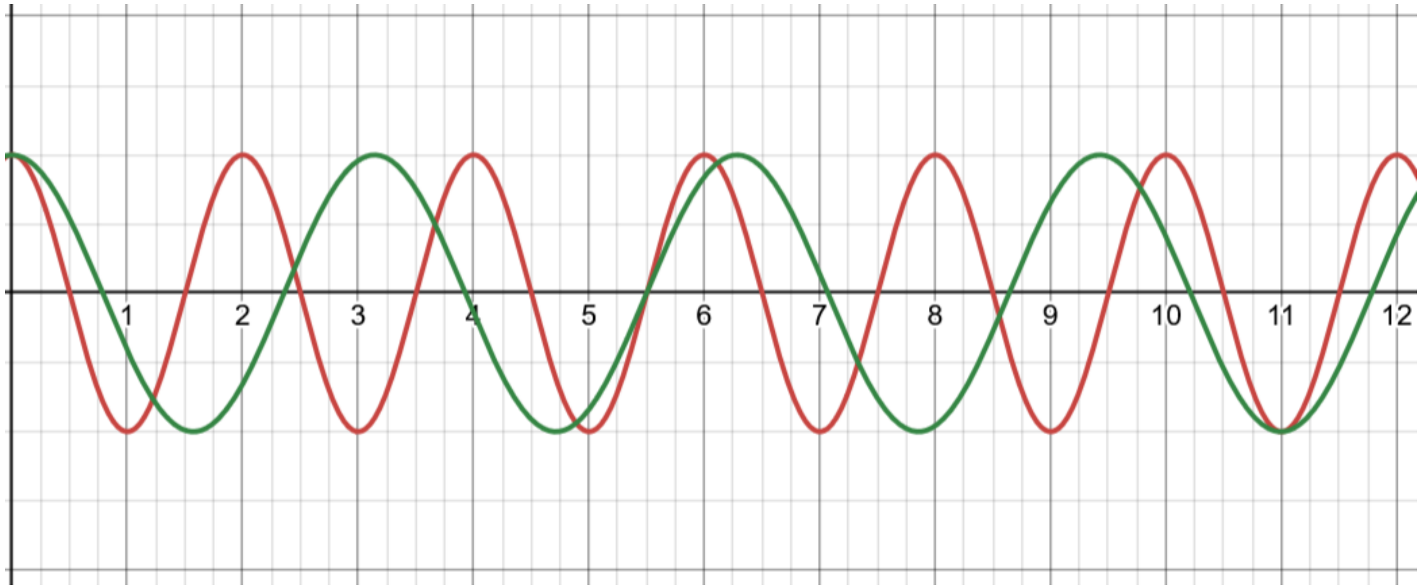

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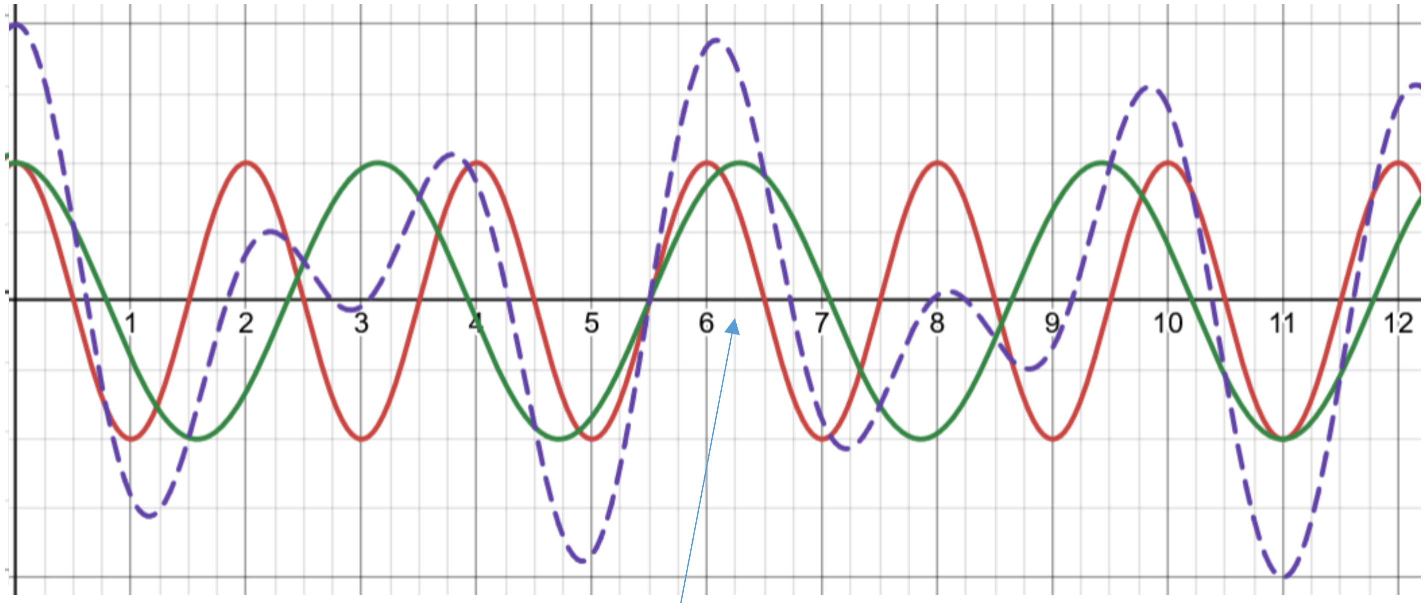

Even at 2π seconds, x_2 would have completed two cycles, while x_1 would have completed three cycles and part of the fourth cycle.

Lesson: sum of two periodic signals is periodic if and only if ratio of their time periods is rational.

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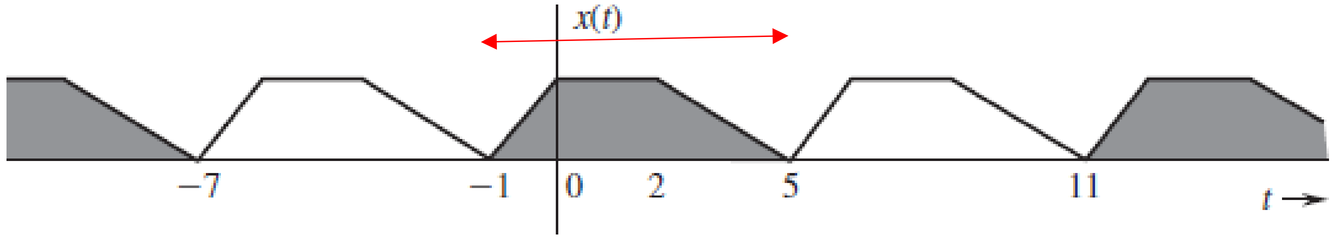
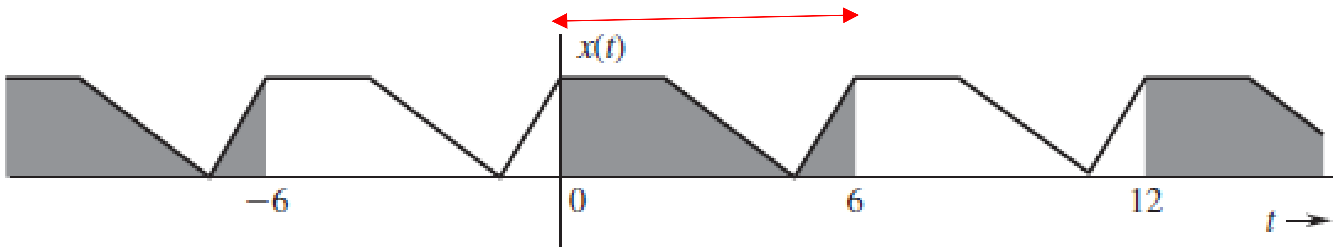
- In this case LCM of their periods exists
- And $T_{sum} = LCM(T_1, T_2)$

Periodic Signal : Interesting Property

Area under one whole period of a periodic signal is always the same no matter where you start!!



(a)



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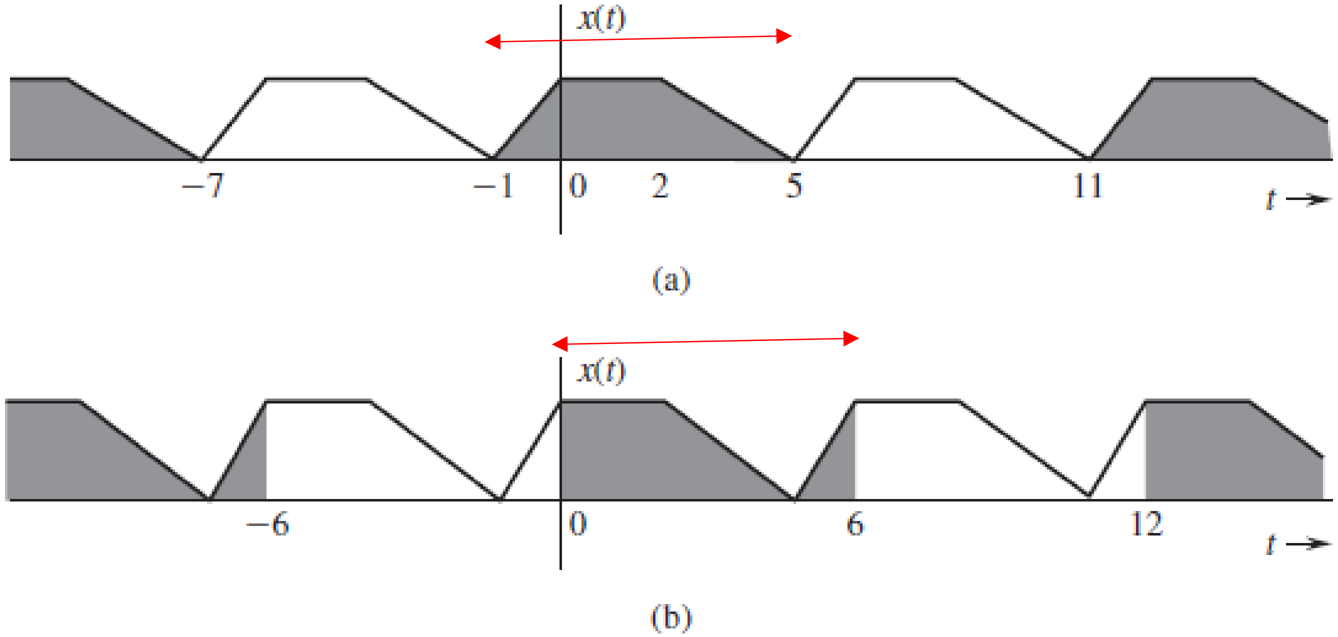
Can you write this mathematically?



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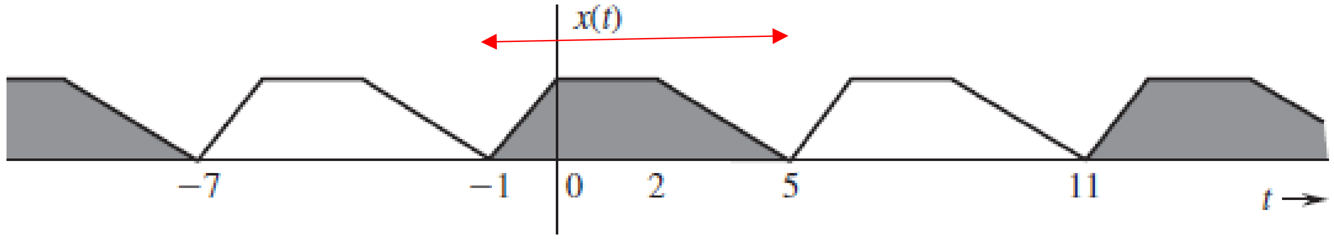

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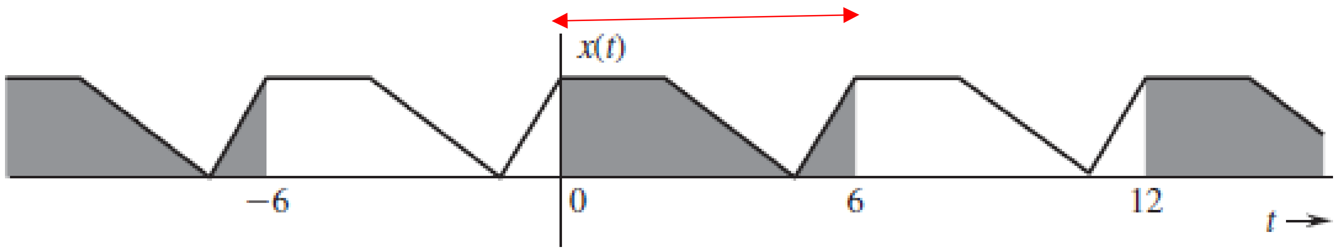

(b)

$$\int_{a}^{a+T_0} x(t)\,dt = \int_{b}^{b+T_0} x(t)\,dt$$

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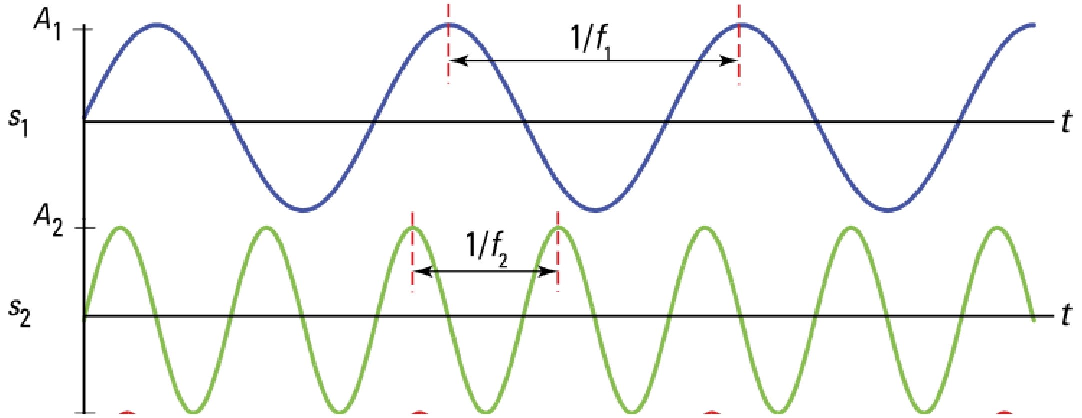
"Deterministic"

Vs.

"Random"



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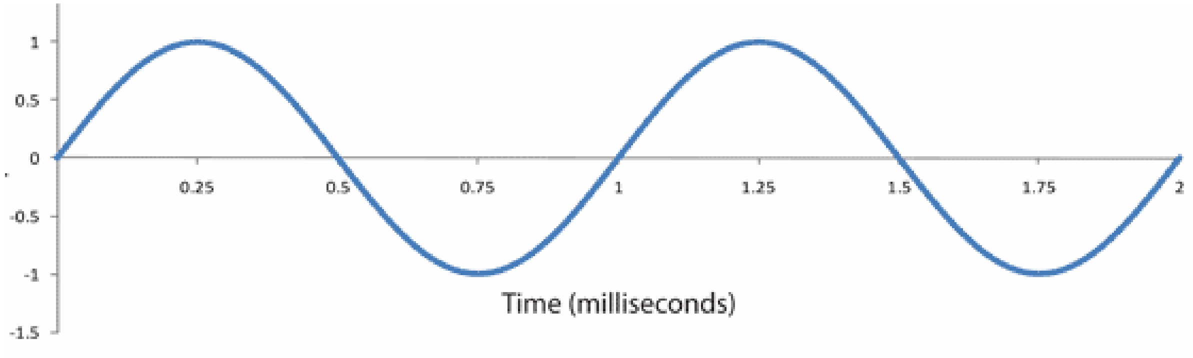

$$x(t) = \sin(2\pi f t) \ \text{ with } f = 1\, kHz$$

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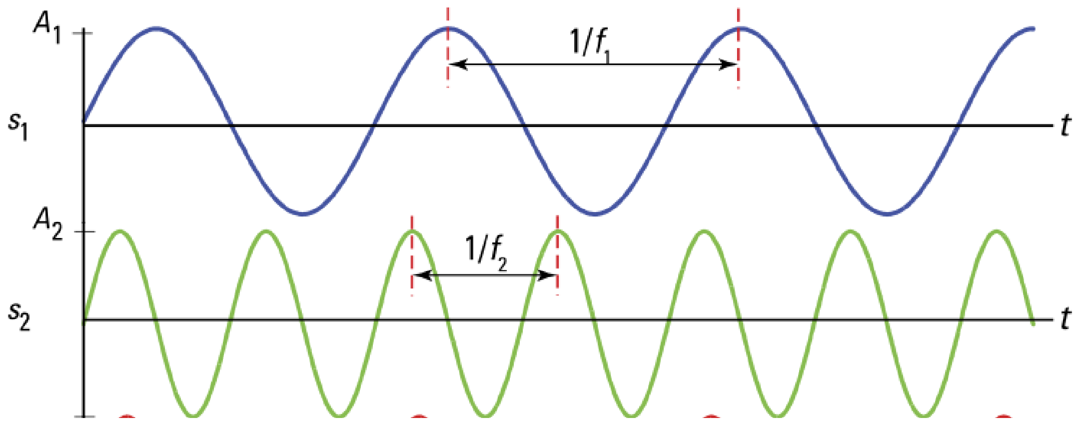

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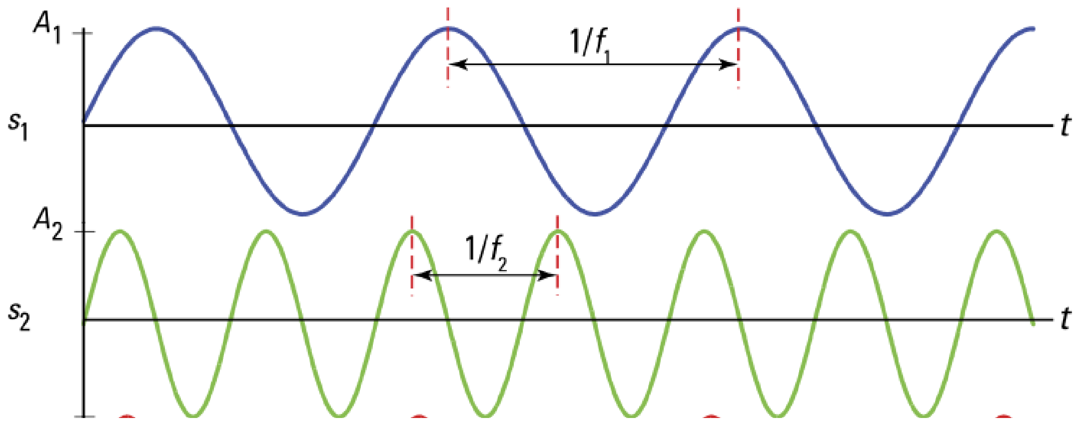

Value of f to be decided by tossing a coin

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Or, we could have random variations in frequency and/or amplitude…

Can you think of a random signal?

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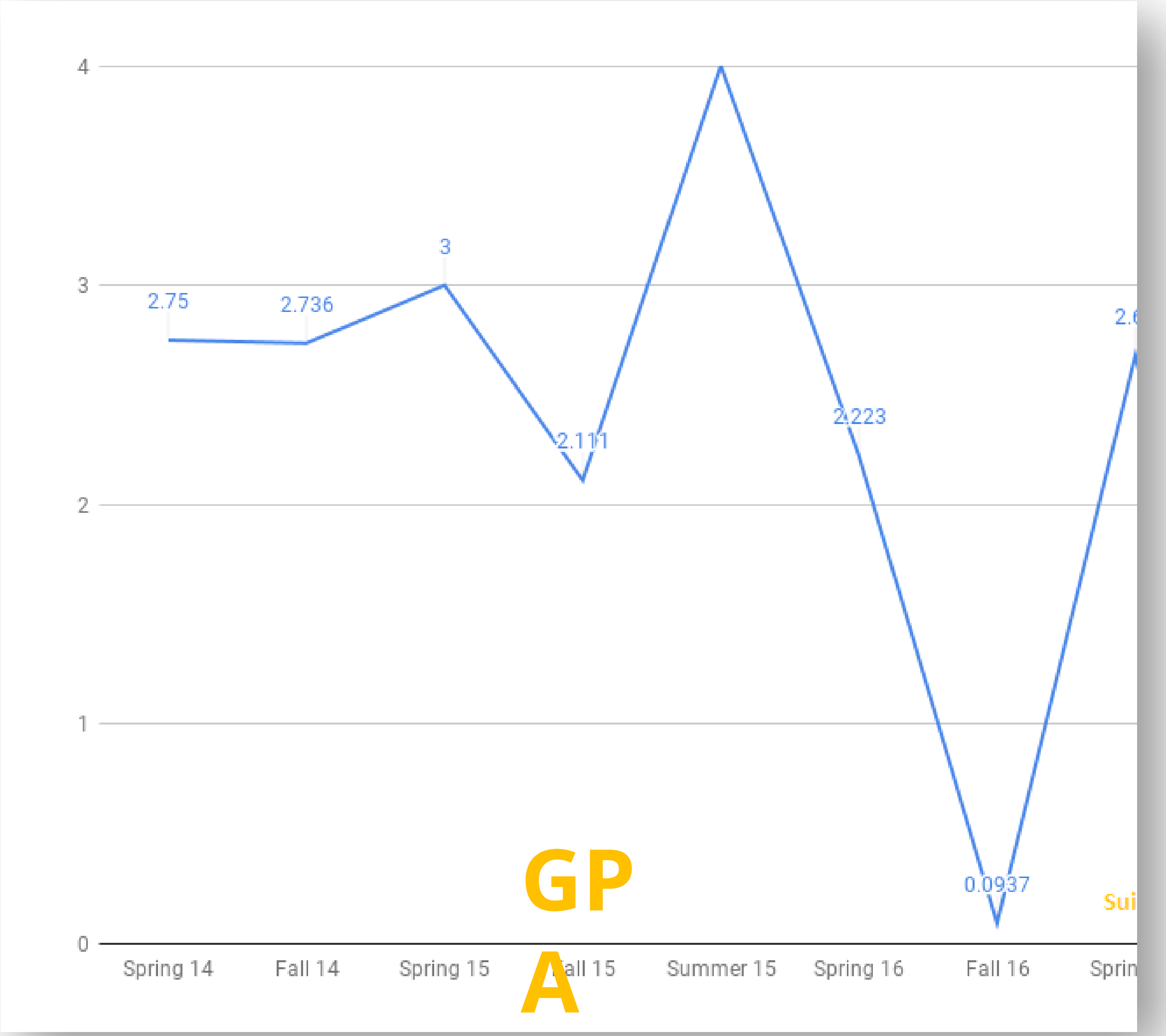

5. Does the signal have finite energy or finite Power?

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For that, we will first need
to answer the following...

- What are energy and power of a signal?
- Why are they needed?

Measures are important…

Height, Age, GDP, Stock Index…

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We often find it useful to look at just one value that gives the overall effect of a signal.

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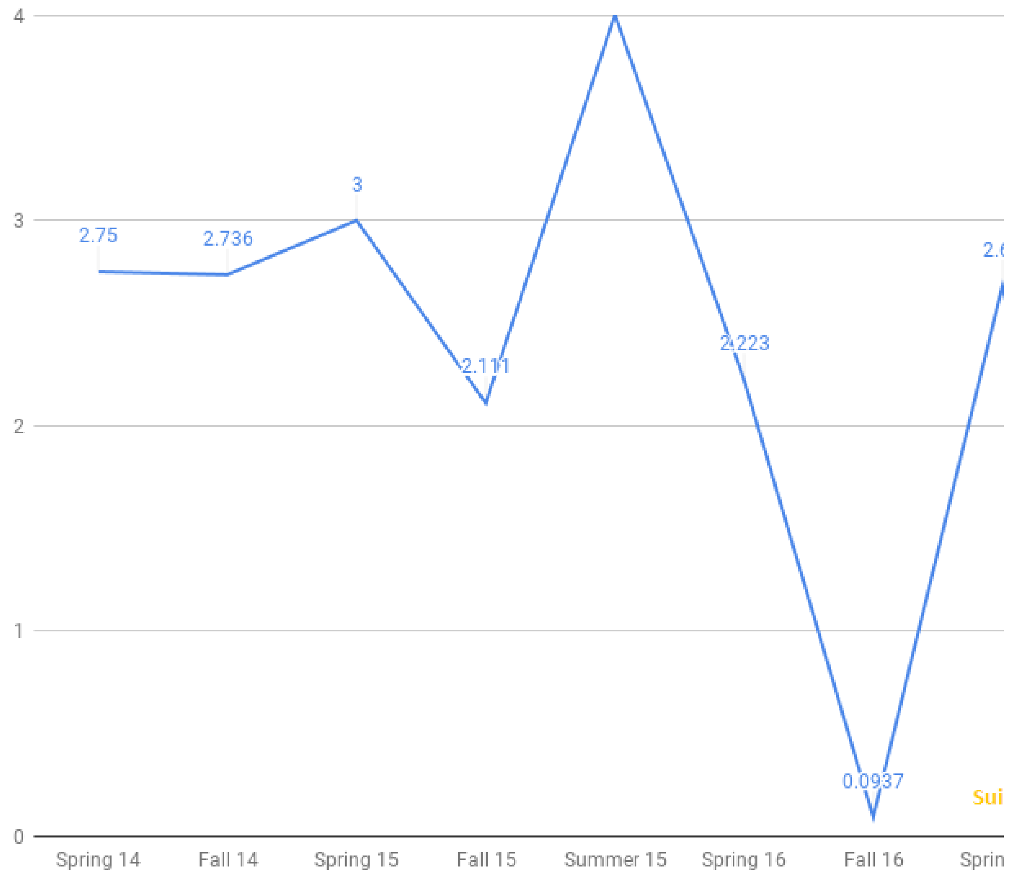

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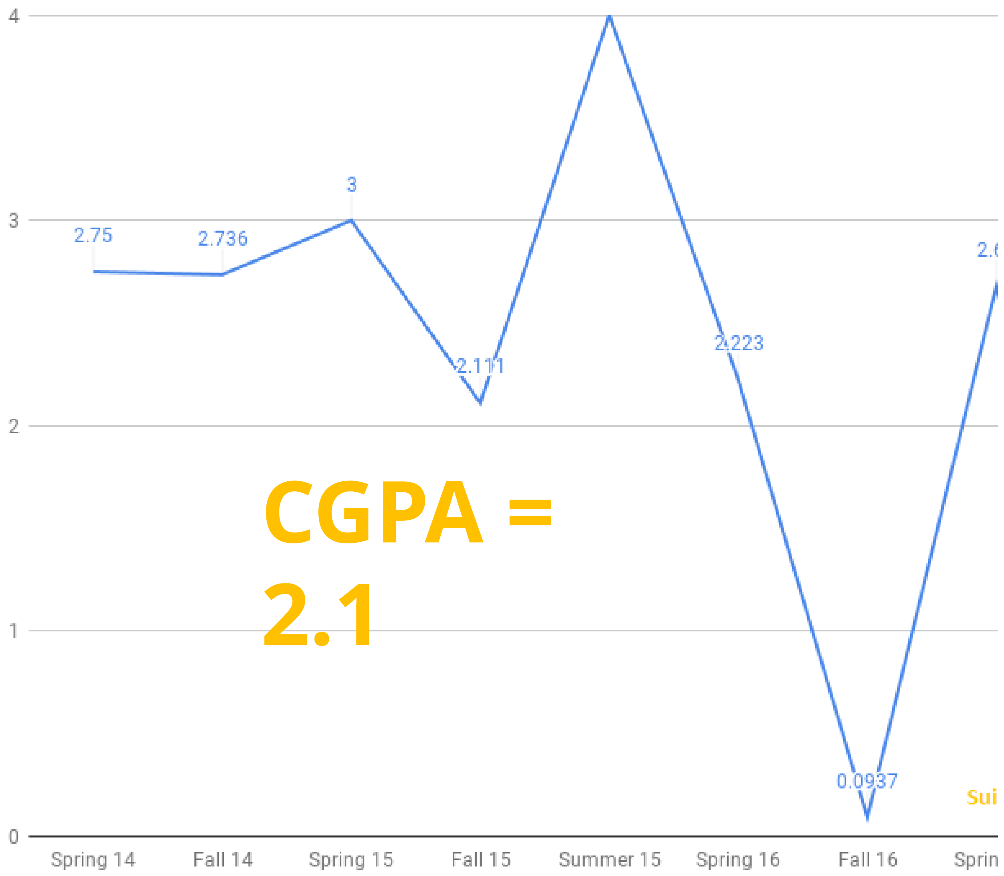

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Two Common Measures of a Signal: Energy & Power

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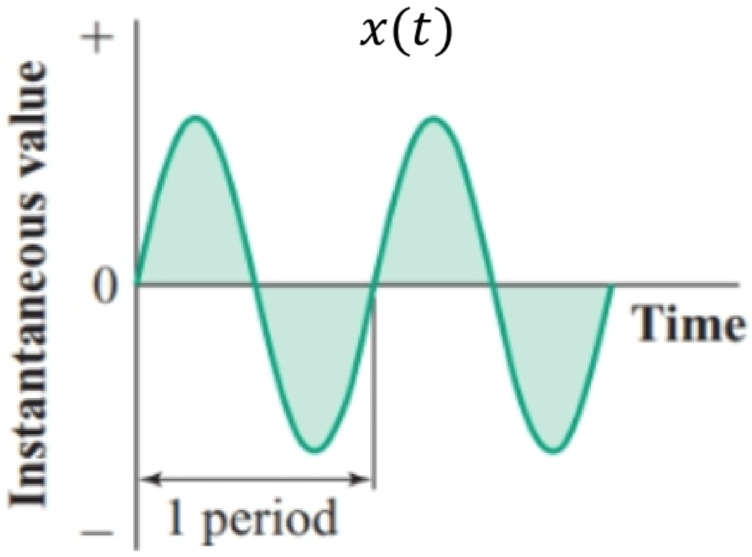

(a) Sine wave

Signal Energy

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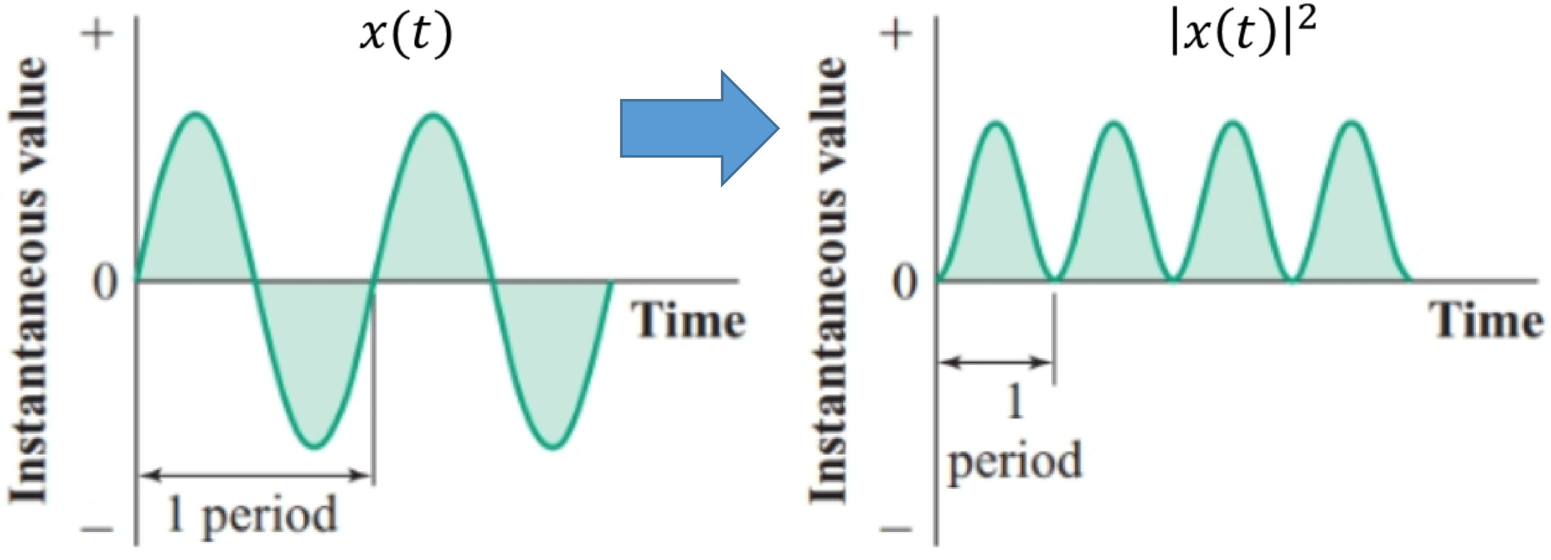

(a) Sine wave (b) Sine-squared wave

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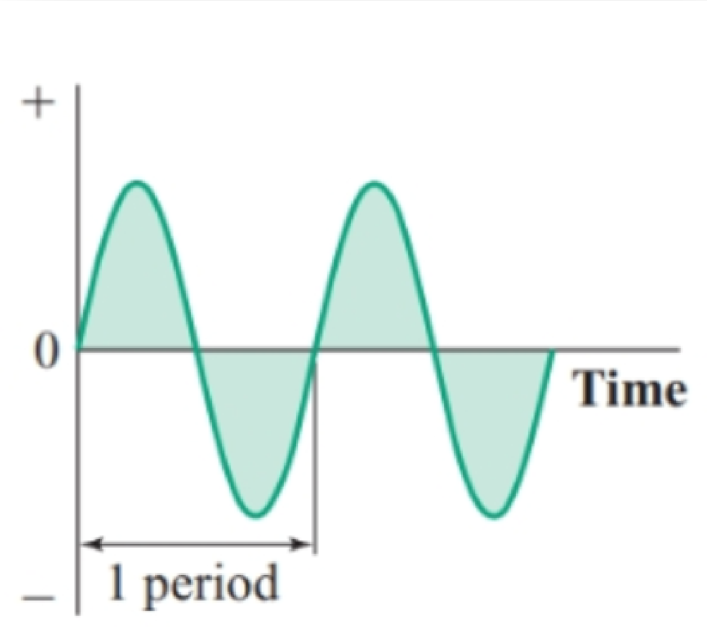

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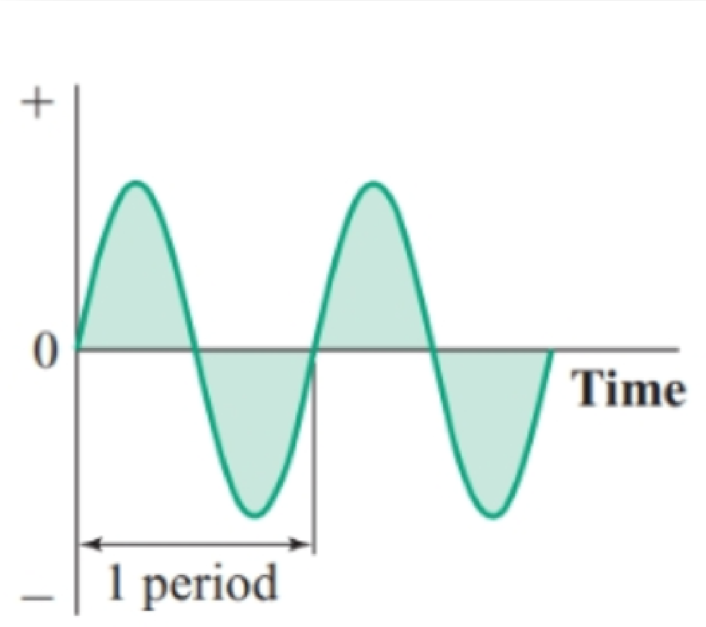

$$\int_{-\infty}^{\infty} x(t) dt = 0$$

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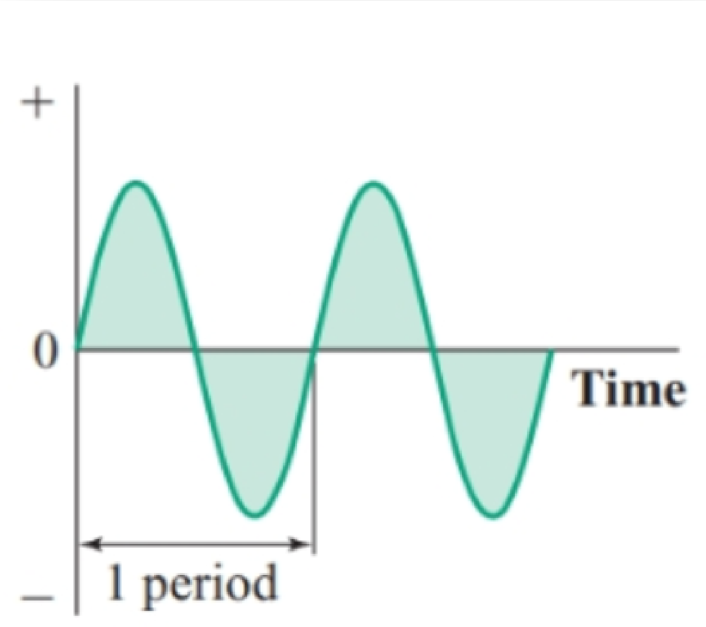

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For many signals, area under the signal curve will turn out to be zero. So not a good way of measuring signals.

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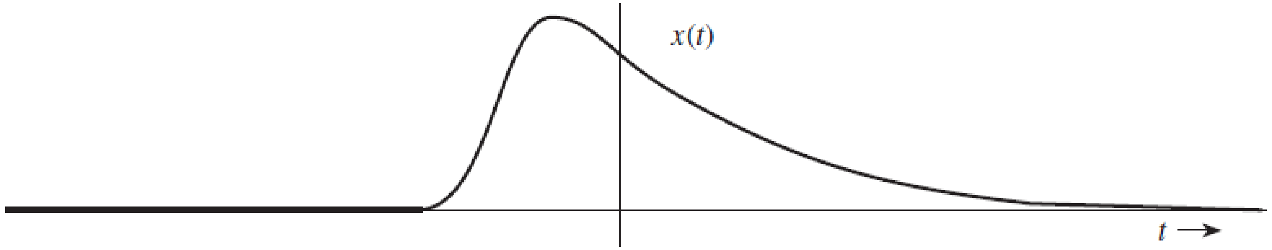
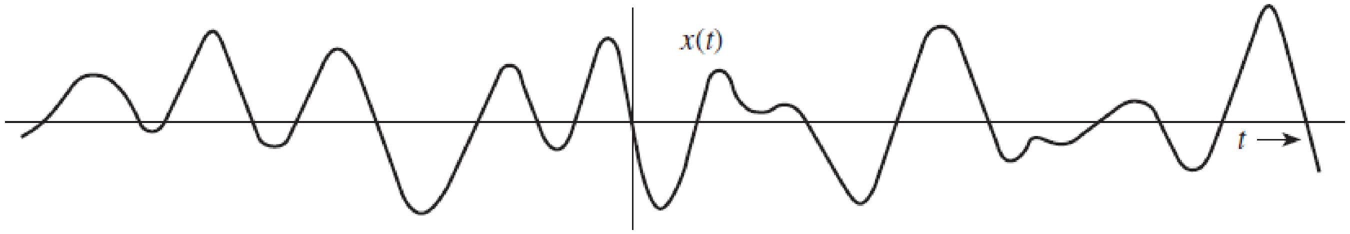
$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 \, dt \neq \infty$$

Q. Which of these have finite energy (assuming $t \to \infty$)?



(a)



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Q. Which of these have finite energy (assuming $t \to \infty$)?

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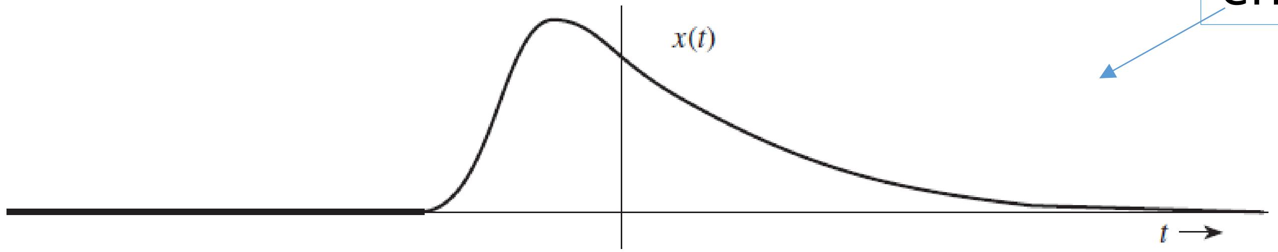

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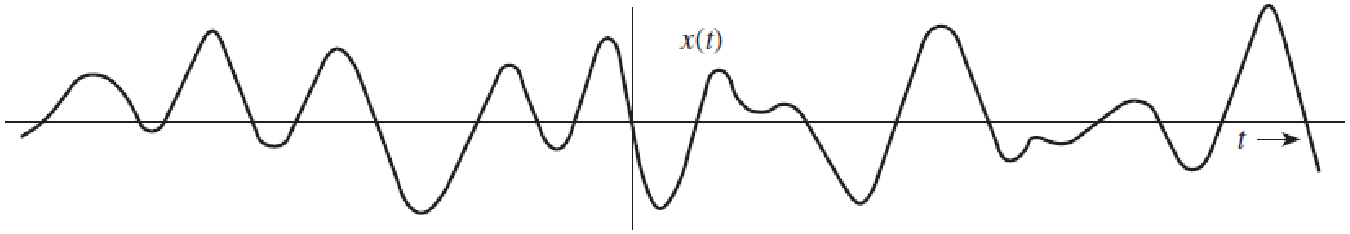

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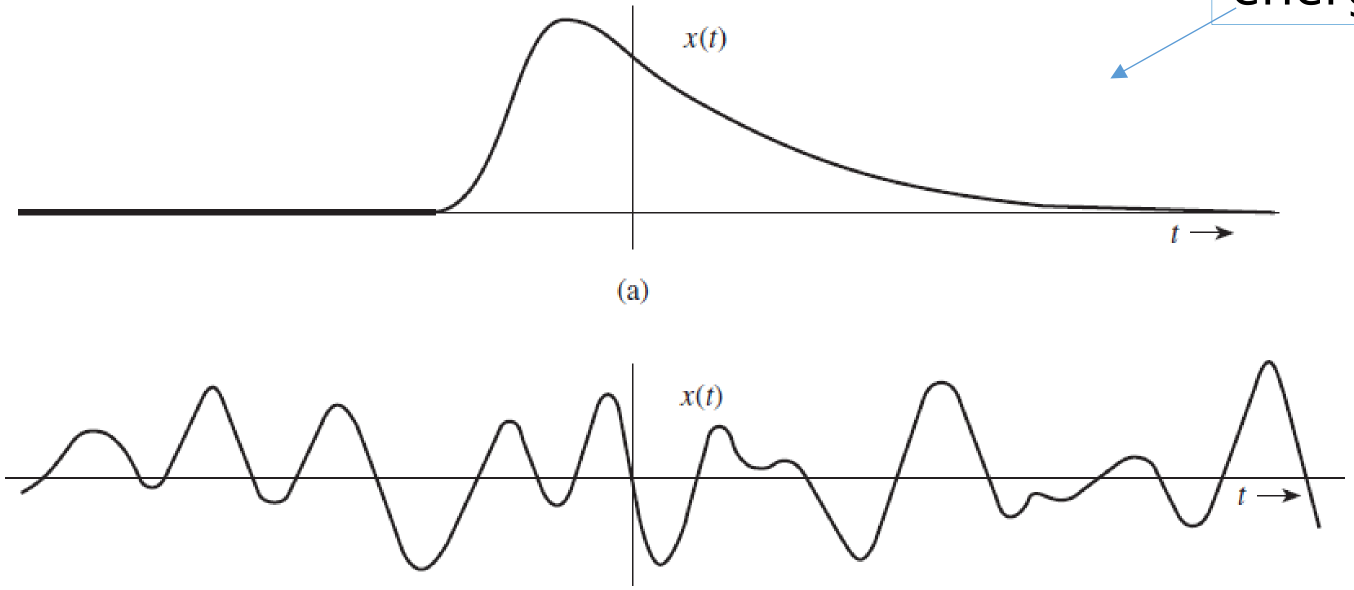

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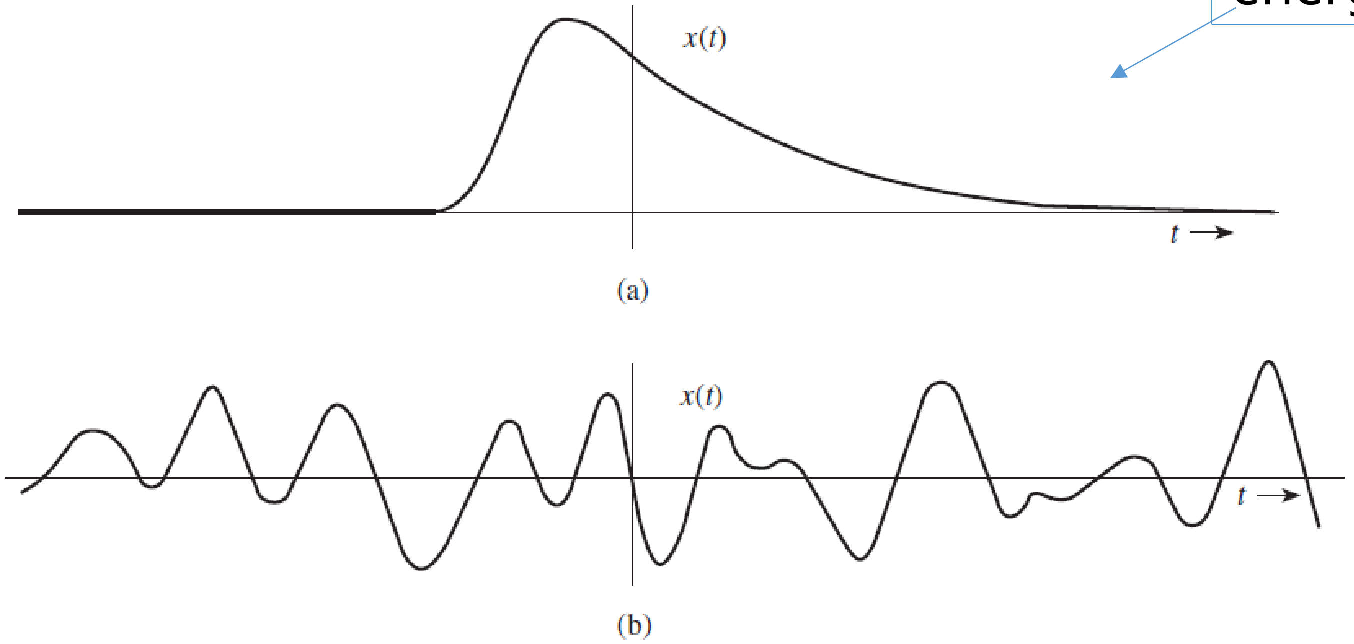

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This is not an "energy signal" as it has infinite energy.

- *So there is no use trying to characterize this signal by its energy!*

Many theoretical signals do not have a finite energy…

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In that case it is more useful to measure their **energy per unit time** (called “Power”)

Signal Power is the signal energy per unit time.

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Mathematically speaking…

$$P_x = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 \, dt$$

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(provided, the limit exists)

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$x(t)$ is a power signal if

$$\boxed{0 < P_x < \infty}$$

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Infinit
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Lesson: an energy signal cannot be a power signal!

- Another measure we sometimes use is the *rms value.*

- rms (root-mean-squared) value of a signal is the square root of its power…

$$x_{rms} = \sqrt{P_x}$$

There are, in fact, many more ways of "measuring" a signal. Try to find some.

HW Activity: Bring to Next Class

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- Show that all practical signals are energy signals (since real-life signals are always time-limited and have no infinite values).

- Show that time-unlimited periodic signals (with no infinite values) are power signals.

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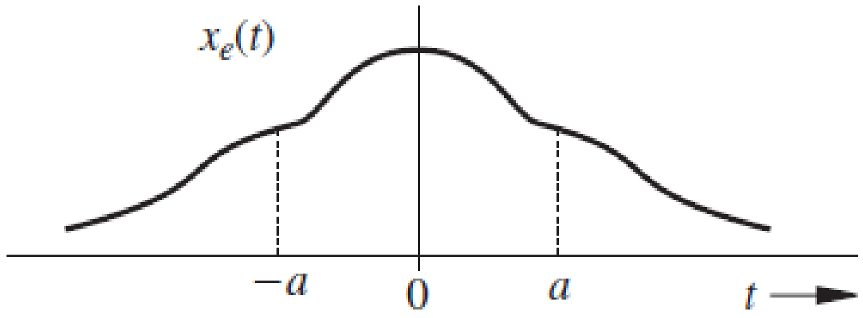

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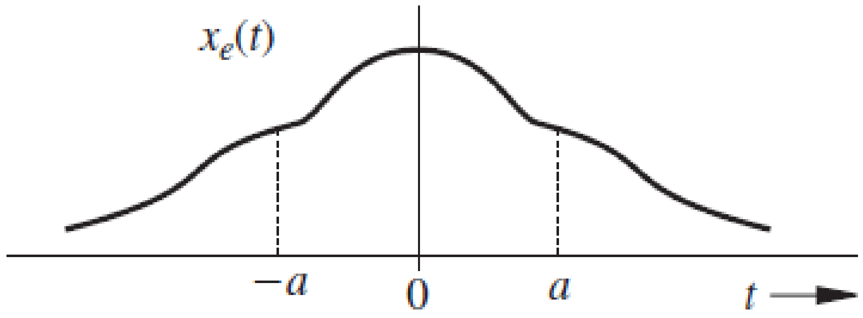

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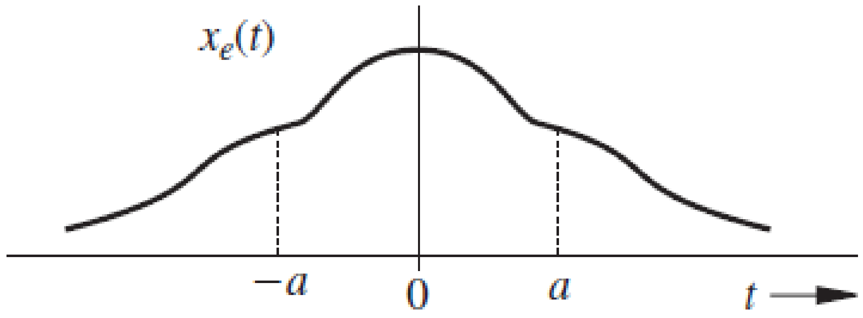

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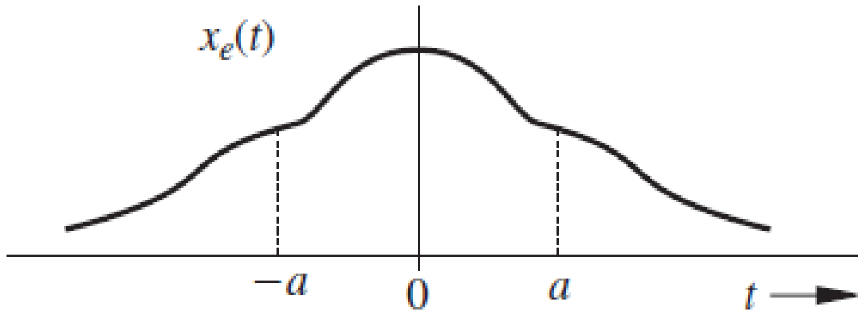

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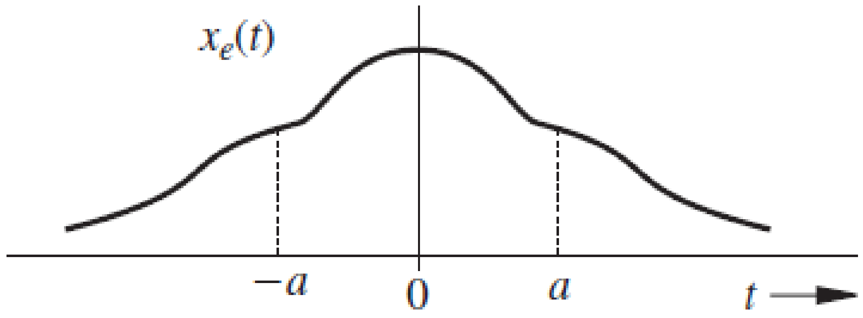

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Even Signal

A signal which is symmetric about the vertical axis is called an even signal.

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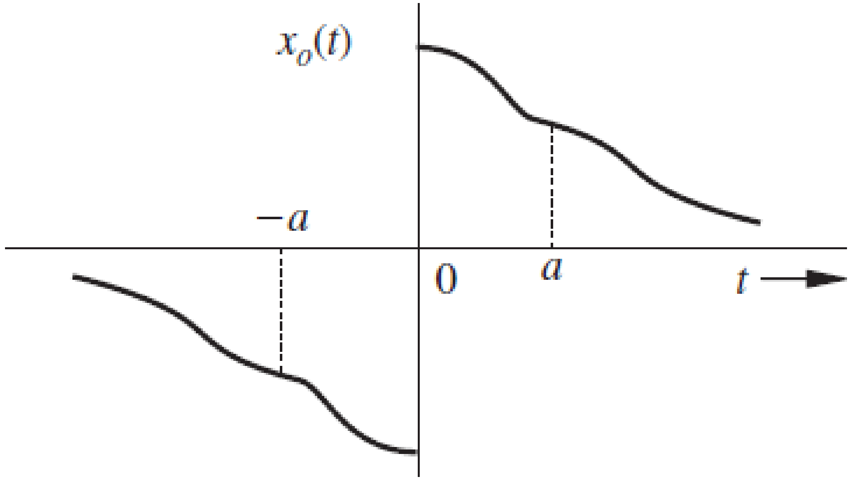

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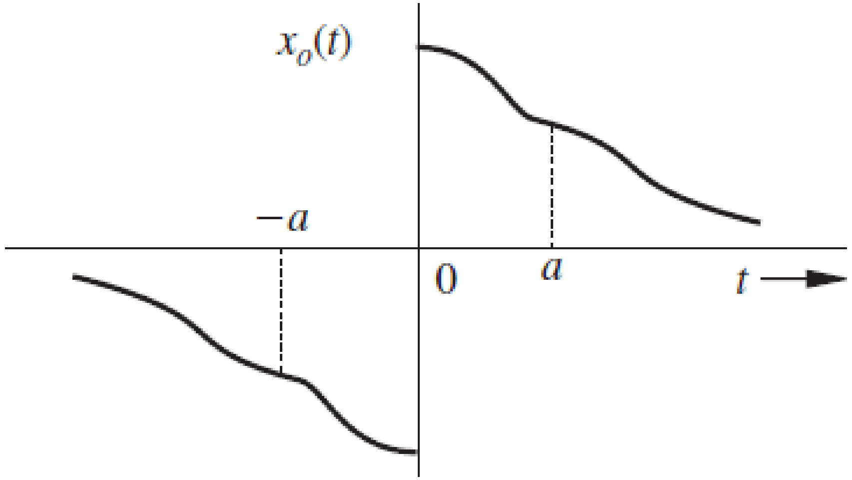

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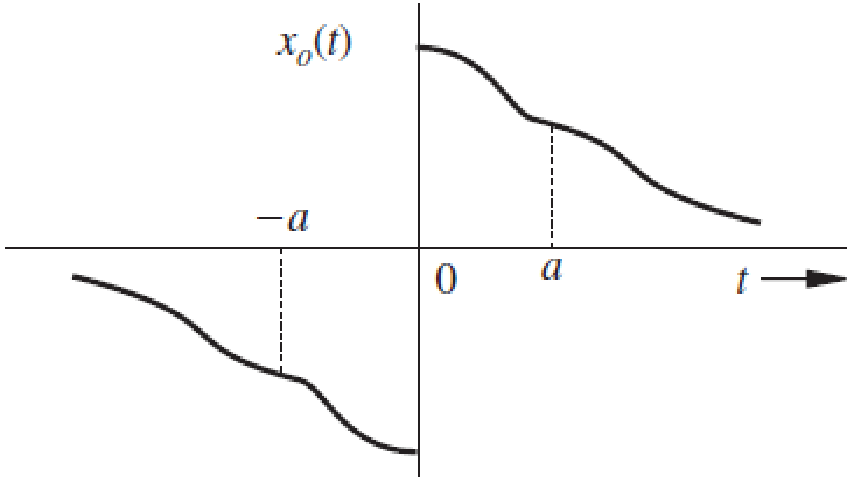

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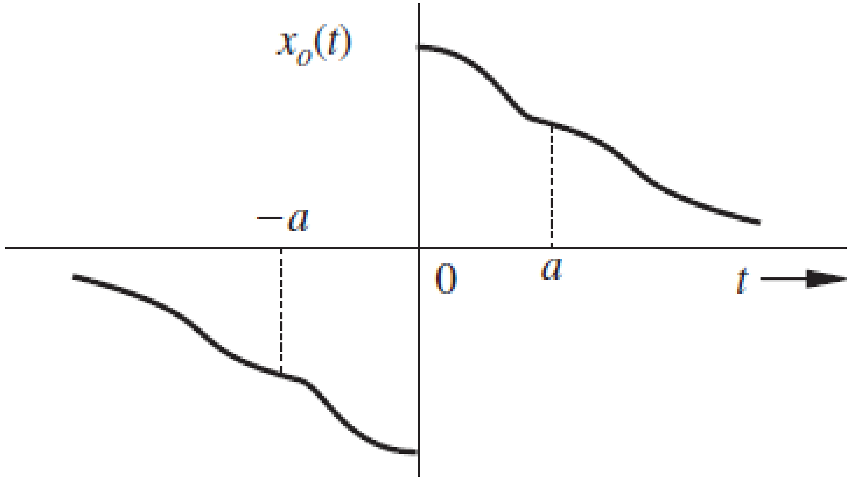

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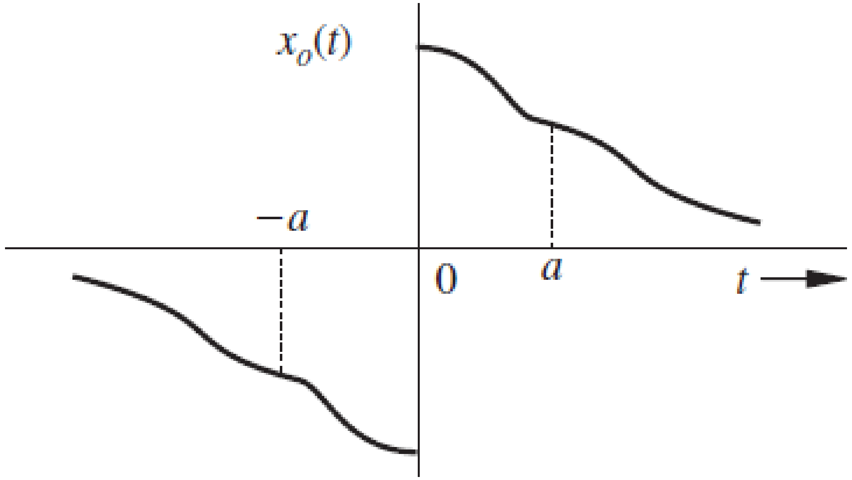

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ODD Signal

A signal which is anti-symmetric about the vertical axis is called an odd signal.

Even vs. Odd

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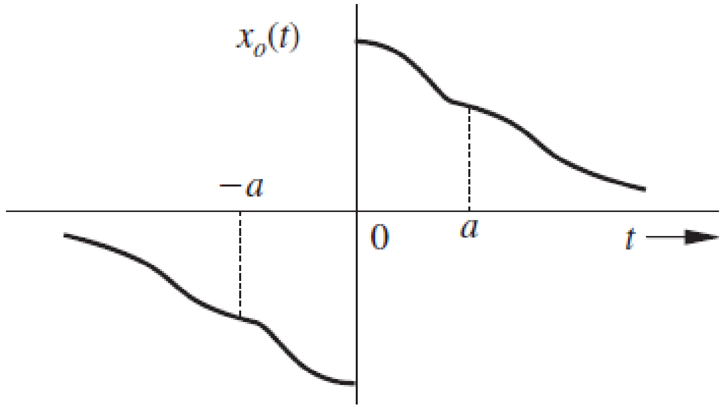
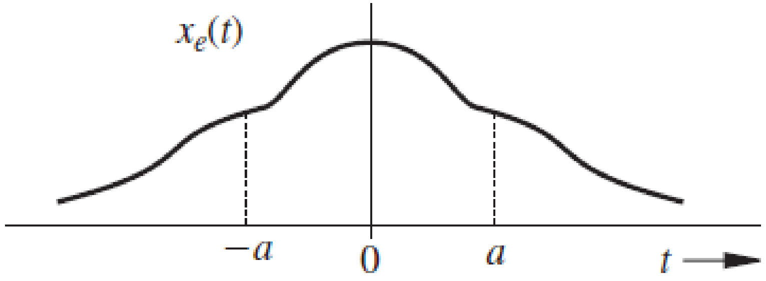


$$\int_{-a}^{a} x_o(t)\,dt$$

Can you simplify these integrals?



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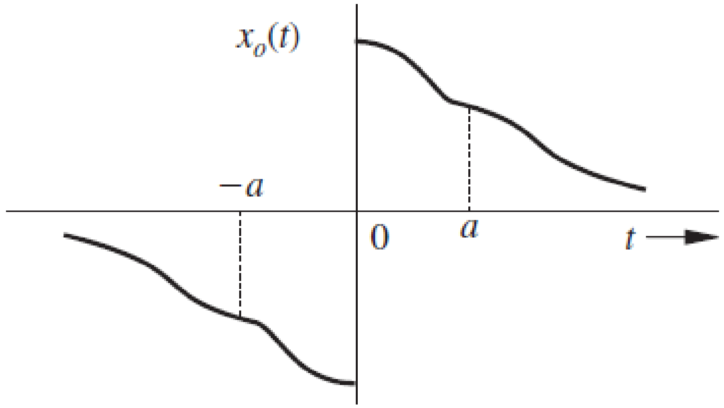

$$\int_{-a}^{a} x_o(t)\,dt = 0$$

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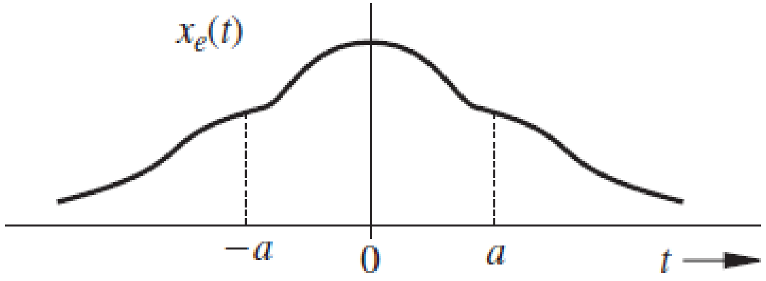

$$\int_{-a}^{a} x_e(t)\,dt = 2\int_{0}^{a} x_e(t)\,dt$$

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Can you think of and draw some functions that are even, odd, or neither?

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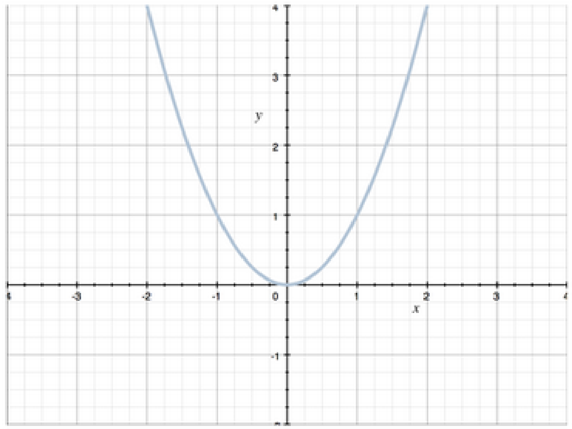
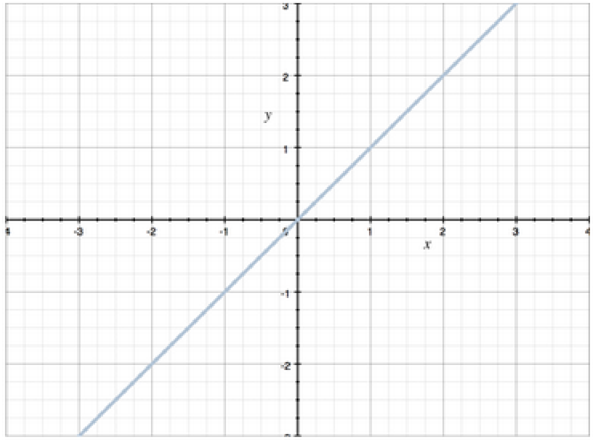
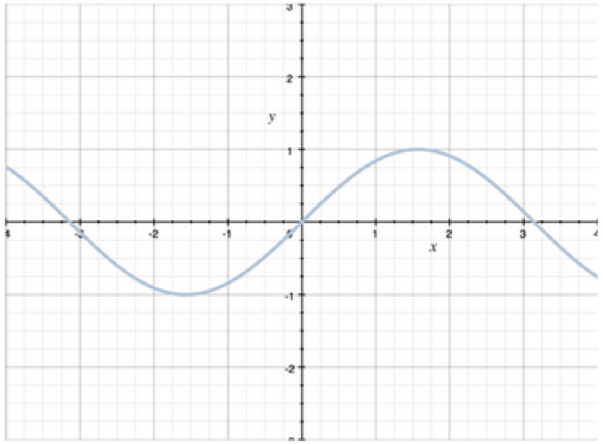
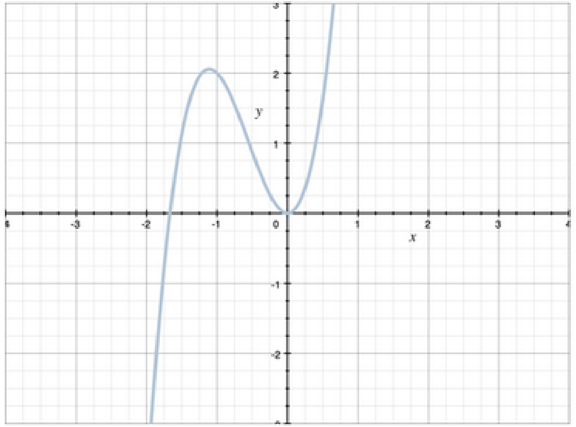
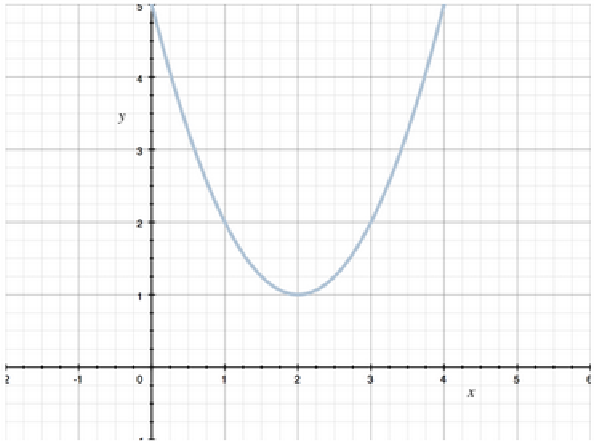
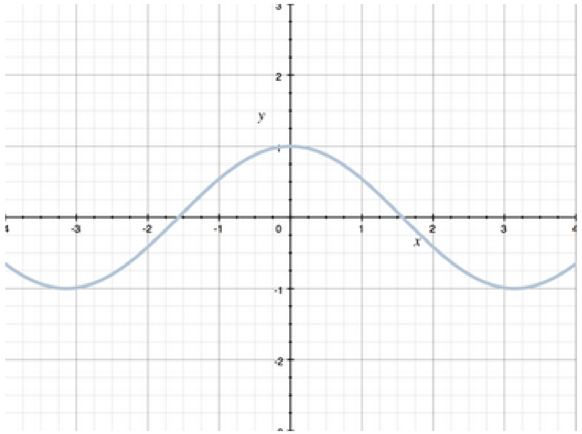






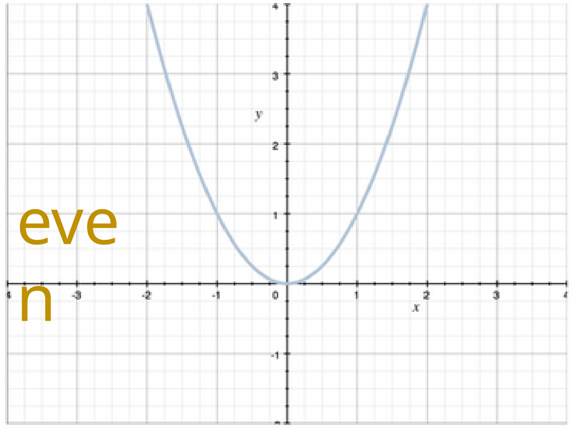


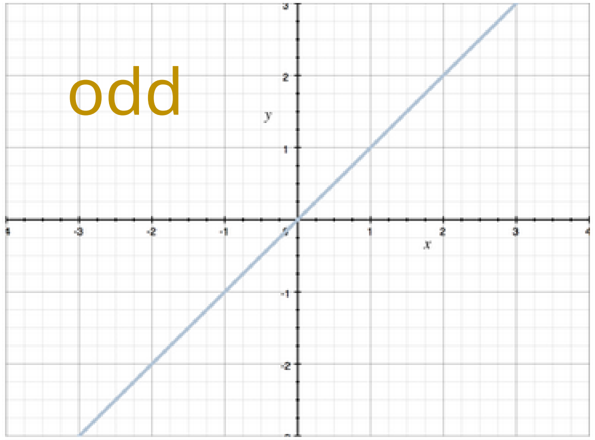


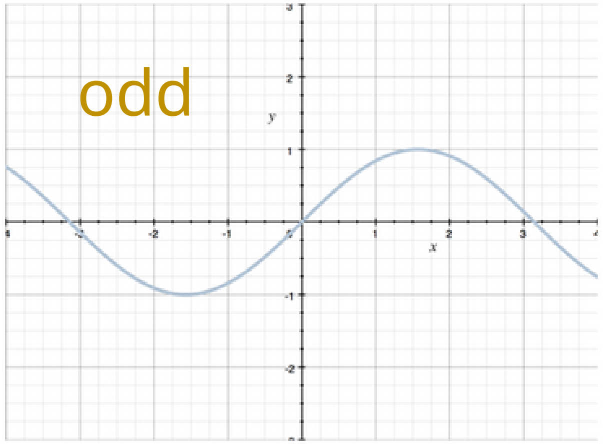


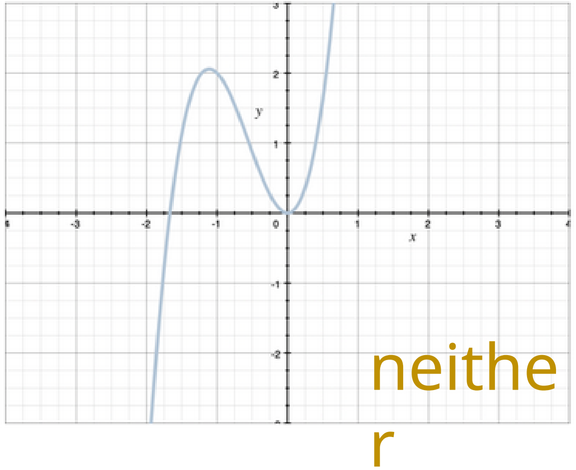
Even vs. Odd

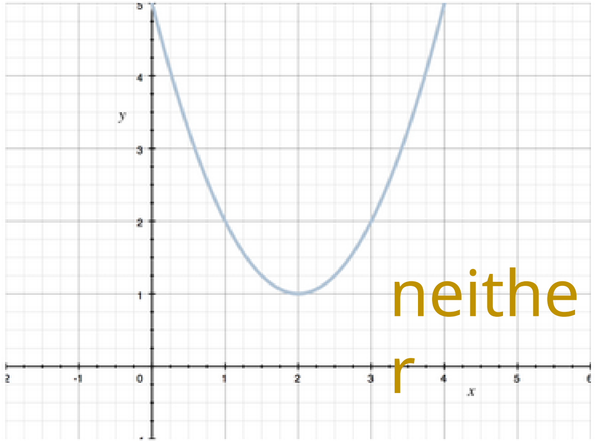
Can you think of and draw some functions that are even, odd, or neither?

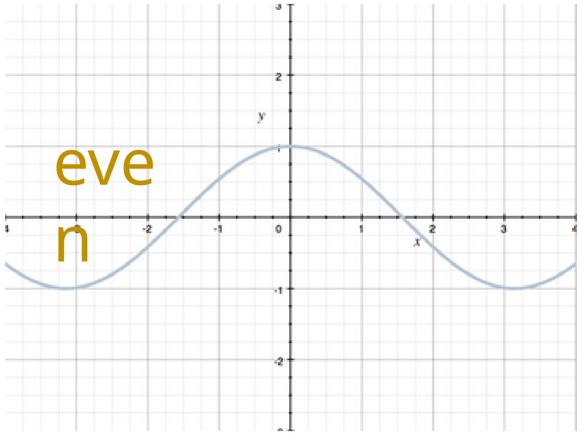












We can write a given function as a sum of even and odd functions

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$$x(t) = \underbrace{\tfrac{1}{2}[x(t) + x(-t)]}_{\text{even}} + \underbrace{\tfrac{1}{2}[x(t) - x(-t)]}_{\text{odd}}$$

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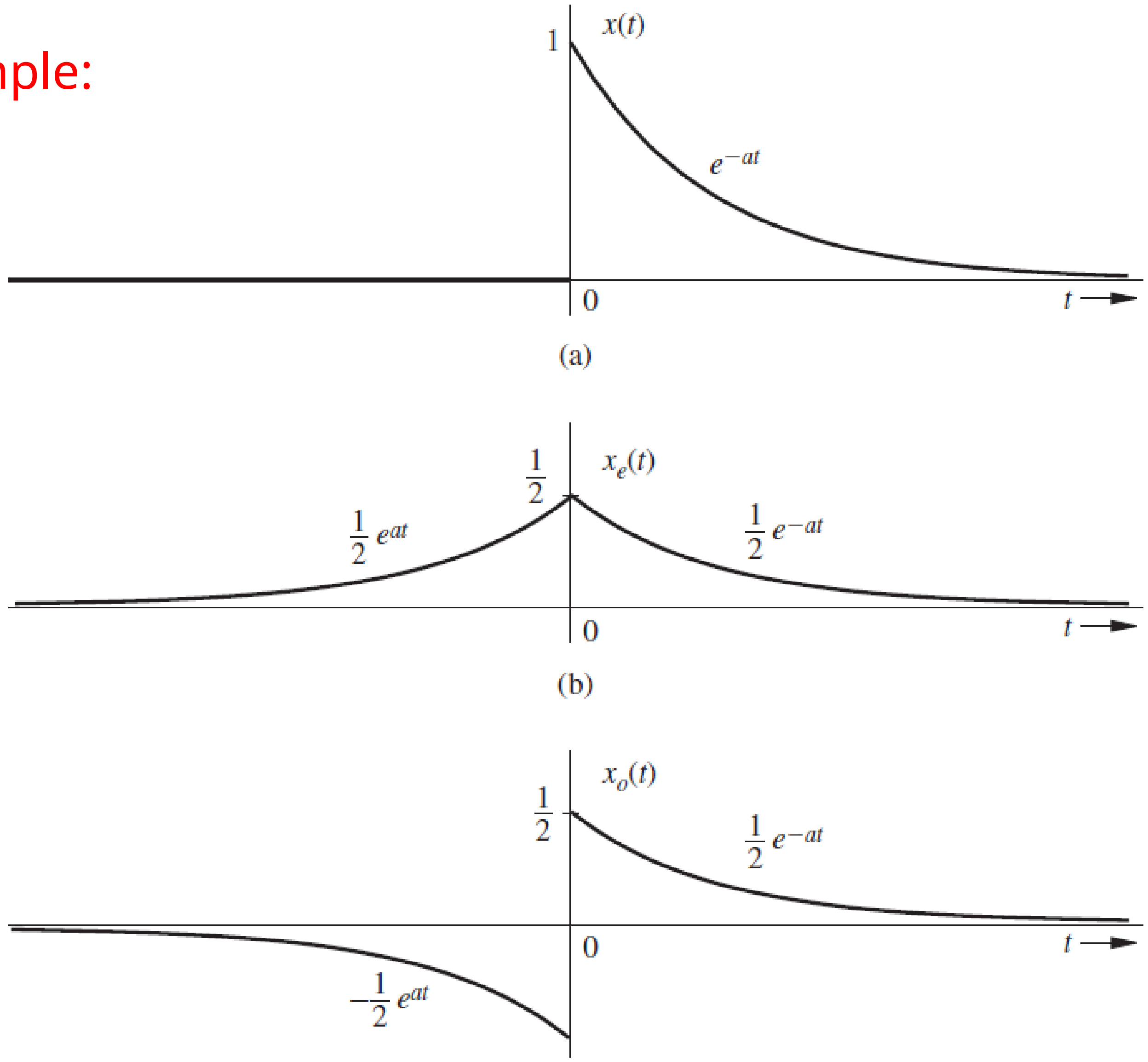
HW: prove that the even and odd parts shown are actually even and odd.

Example:



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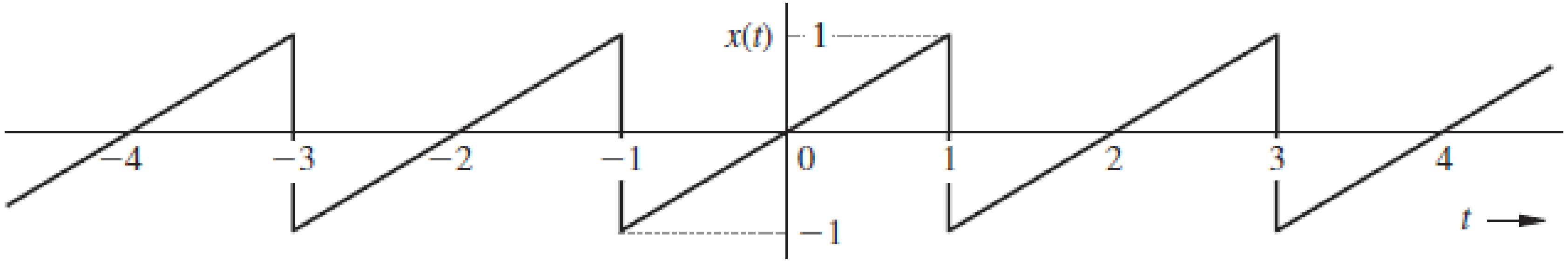
Let us analyze some more signals…



Consider the signal shown above (with $t \to \infty$):

- Is it periodic? If so, find the period.
- Is it an energy signal?
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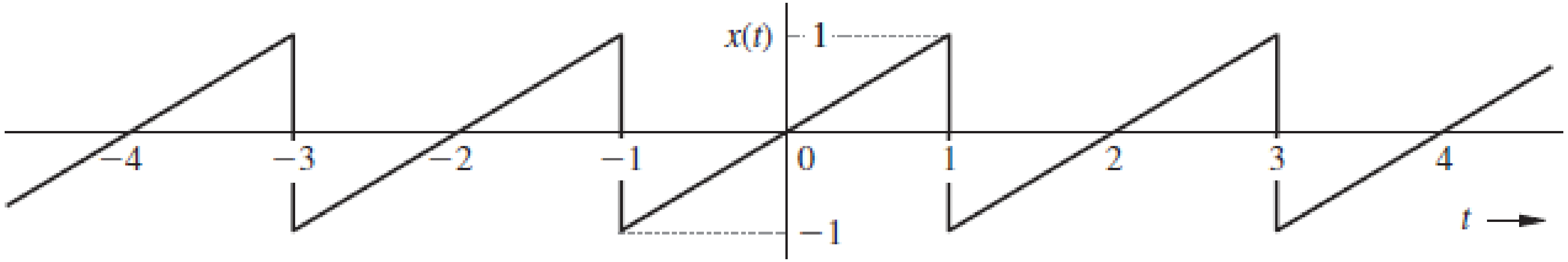

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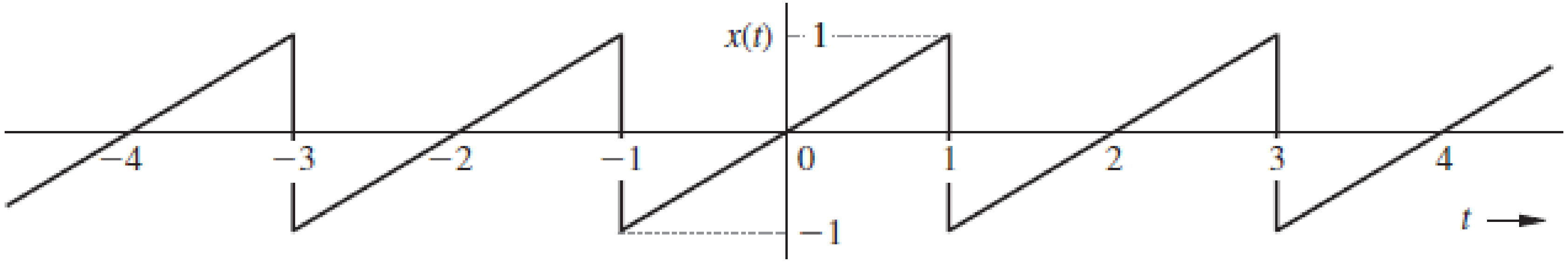
- Yes, the signal is periodic with period $T_0 = 2$



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- Yes, the signal is periodic with period $T_0 = 2$
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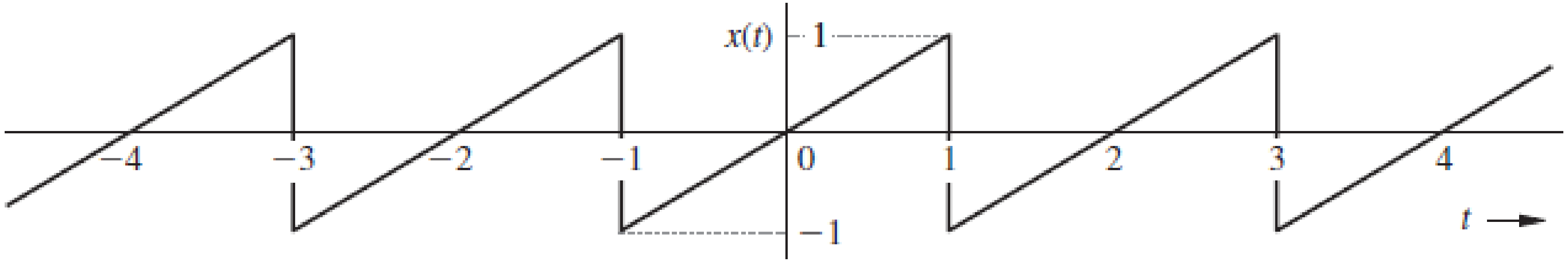
  - Why? Clearly, the absolute square of the signal will have infinite area under it.



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- Yes, the signal is periodic with period $T_0 = 2$
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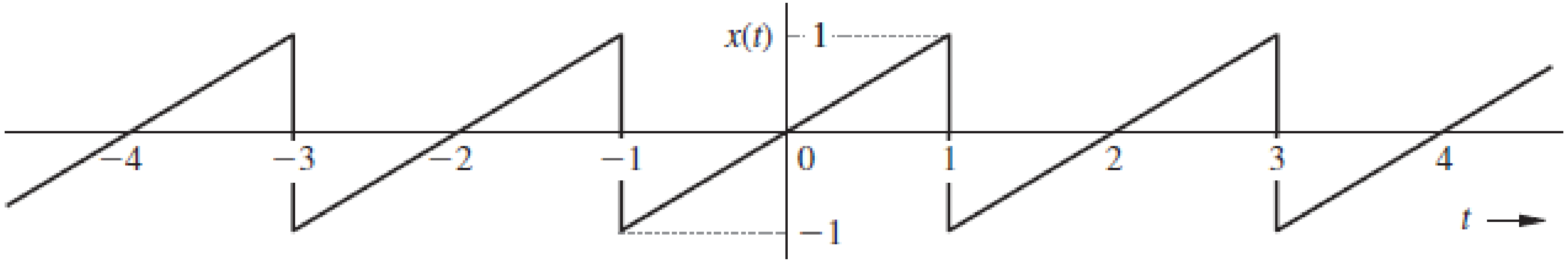
- It is a power signal as it's power $P_x = \frac{1}{3}$ is neither zero nor infinite.



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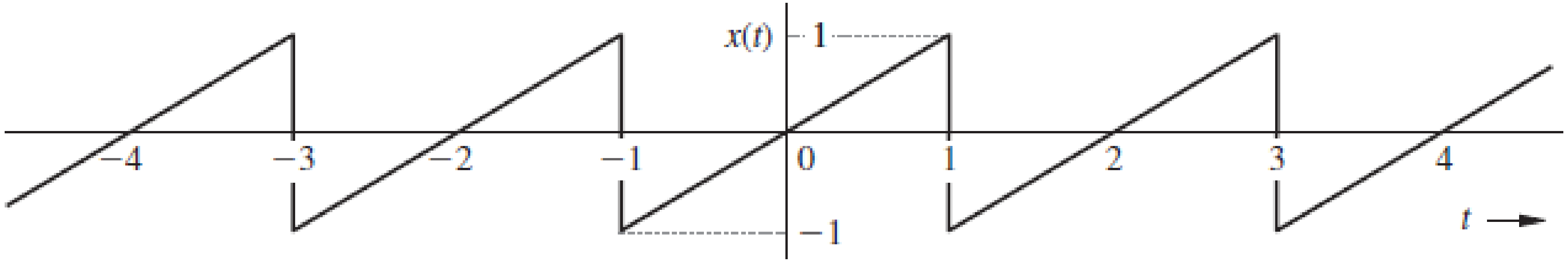

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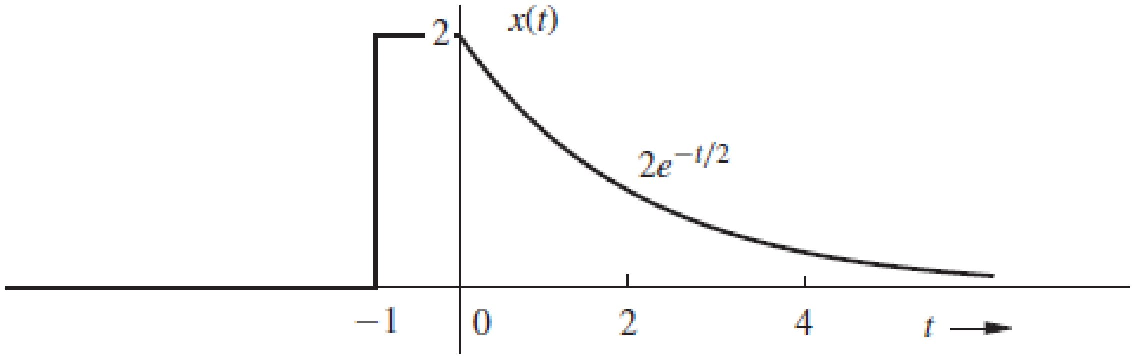
Here, we have used the formula for the power of a periodic signal and that fact that given signal's period from -1 to 1 can be modeled as $x(t) = t$



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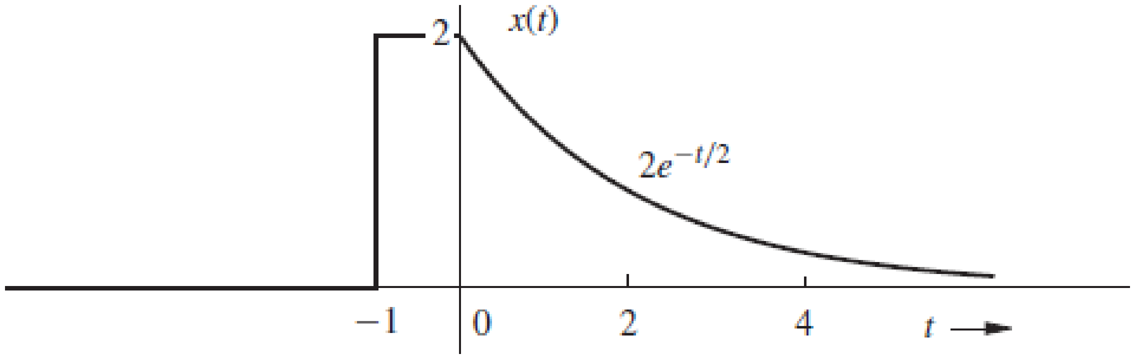

First, can you write this signal mathematically?

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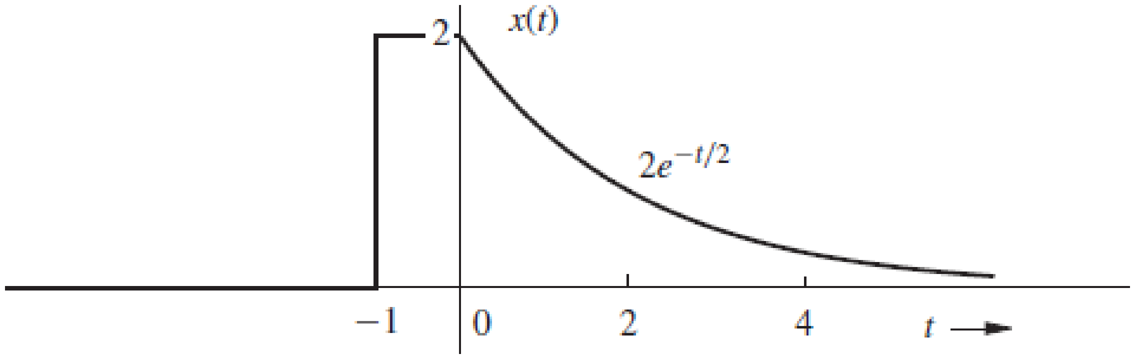
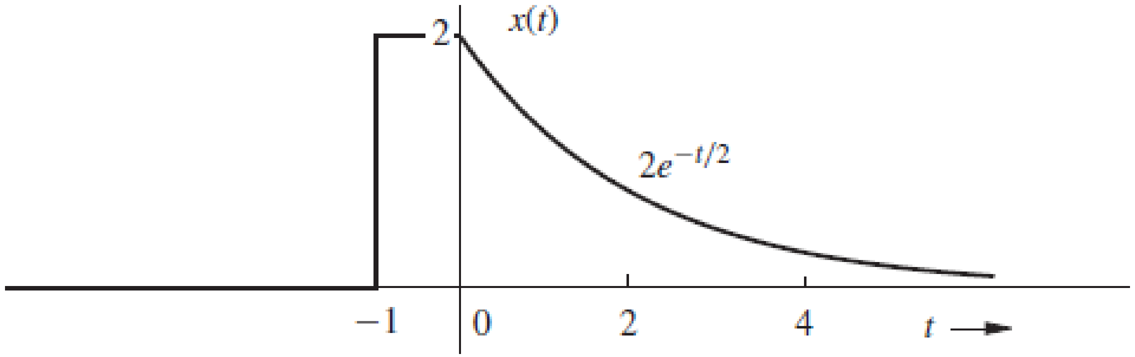


Consider the signal shown above (with $t \to \infty$):

- Is it periodic? If so, find the period.
- Is it an energy signal?
- Is it a power signal?

First, can you write this signal mathematically?

$$x(t) = \begin{cases} 2, & -1 \leq t \leq 0 \\ 2e^{-t/2}, & t \geq 0 \end{cases}$$



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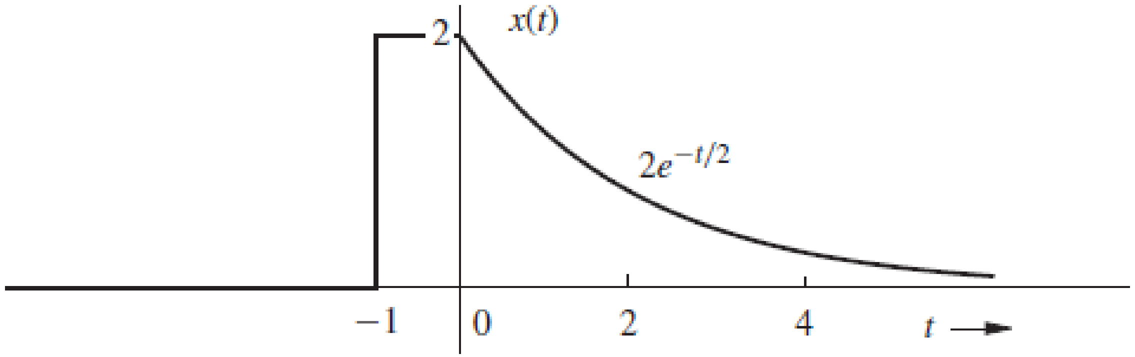
- Is it a power signal?

- It is not periodic.



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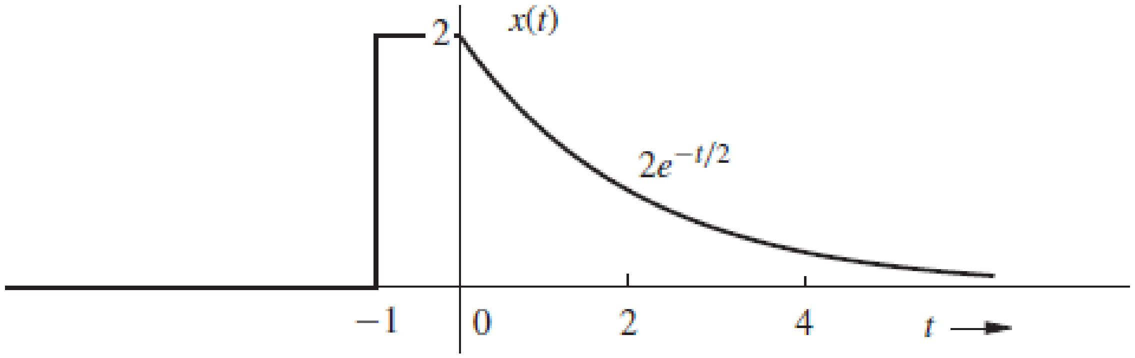
$$x(t) = \begin{cases} 2, & -1 \le t \le 0 \\ 2e^{-t/2}, & t \ge 0 \end{cases}$$

Consider the signal shown above (with $t \to \infty$):

- Is it periodic? If so, find the period.
- Is it an energy signal?
- Is it a power signal?

- It is not periodic.

- It is an energy signal since it energy $E_x = 8$ is finite.



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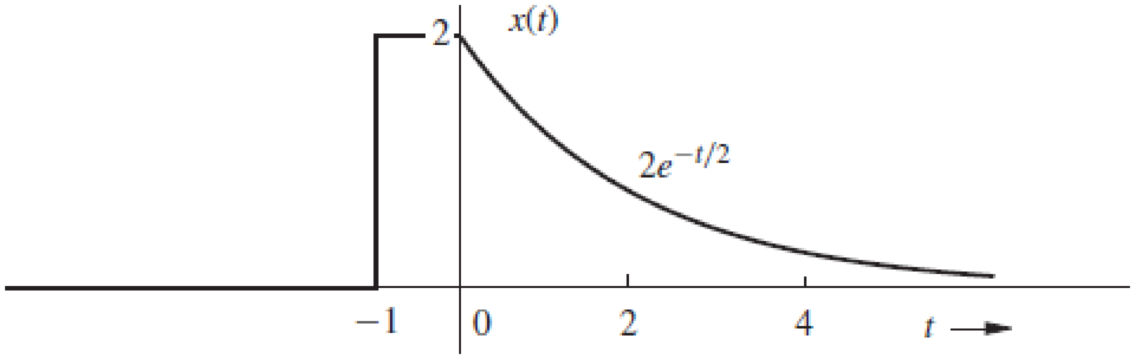
$$E_x = \int_{-\infty}^{\infty} |x(t)|^2\,dt = \int_{-1}^{0} (2)^2\,dt + \int_{0}^{\infty} 4e^{-t}\,dt = 4 + 4 = 8$$



First, can you write this signal mathematically?

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Consider the signal shown above (with $t \to \infty$):

- Is it periodic? If so, find the period.
- Is it an energy signal?
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$$E_x = \int_{-\infty}^{\infty} |x(t)|^2\, dt = \int_{-1}^{0} (2)^2\, dt + \int_{0}^{\infty} 4e^{-t}\, dt = 4 + 4 = 8$$

- It is not periodic.
- It is an energy signal since it energy $E_x = 8$ is finite.
- It is not a power signal, since an energy signal will always have zero power.

Questions?? Thoughts??